

SECTION 32 12 16
ASPHALT PAVING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Cold milling of existing hot-mix asphalt pavement.
2. Hot-mix asphalt patching.
3. Hot-mix asphalt paving.
4. Hot-mix asphalt paving overlay.
5. Pavement-marking paint.

B. Related Sections:

1. Division 31 Section "Earth Moving" for aggregate subbase and base courses and for aggregate pavement shoulders.
2. Division 32 Section "Concrete Paving Joint Sealants" for joint sealants and fillers at paving terminations.

1.2 SUBMITTALS

A. Product Data: For each type of product indicated. Include technical data and tested physical and performance properties.

1. Job-Mix Designs: Certification, by authorities having jurisdiction, of approval of each job mix proposed for the Work.
2. Job-Mix Designs: For each job mix proposed for the Work.

B. Material Certificates: For each paving material, from manufacturer.

1.3 QUALITY ASSURANCE

A. Manufacturer Qualifications: A paving-mix manufacturer registered with and approved by authorities having jurisdiction or the DOT of state in which Project is located.

B. Regulatory Requirements: Comply with materials, workmanship, and other applicable requirements of Nye County Public Works and/or Nevada Department of Transportation for asphalt paving work.

1. Measurement and payment provisions and safety program submittals included in standard specifications do not apply to this Section.

C. Preinstallation Conference: Conduct conference at Tonopah Elementary School.

1.4 PROJECT CONDITIONS

- A. Environmental Limitations: Do not apply asphalt materials if subgrade is wet or excessively damp, if rain is imminent or expected before time required for adequate cure, or if the following conditions are not met:
 - 1. Tack Coat: Minimum surface temperature of 60 deg F.
 - 2. Asphalt Base Course: Minimum surface temperature of 40 deg F and rising at time of placement.
 - 3. Asphalt Surface Course: Minimum surface temperature of 60 deg F at time of placement.
- B. Pavement-Marking Paint: Proceed with pavement marking only on clean, dry surfaces and at a minimum ambient or surface temperature of 40 deg F for oil-based materials, 55 deg F water-based materials, and not exceeding 95 deg F.

PART 2 - PRODUCTS

2.1 AGGREGATES

- A. Coarse Aggregate: ASTM D 692, sound; angular crushed stone, crushed gravel, or cured, crushed blast-furnace slag.
- B. Fine Aggregate: ASTM D 1073 or AASHTO M 29, sharp-edged natural sand or sand prepared from stone, gravel, cured blast-furnace slag, or combinations thereof.
- C. Mineral Filler: ASTM D 242 or AASHTO M 17, rock or slag dust, hydraulic cement, or other inert material.

2.2 ASPHALT MATERIALS

- A. Asphalt Binder: AASHTO M 320 or AASHTO MP 1a, PG 64-22 or PG 76-22.
- B. Tack Coat: ASTM D 977 or AASHTO M 140 emulsified asphalt, or ASTM D 2397 or AASHTO M 208 cationic emulsified asphalt, slow setting, diluted in water, of suitable grade and consistency for application.

2.3 AUXILIARY MATERIALS

- A. Herbicide: Commercial chemical for weed control, registered by the EPA. Provide in granular, liquid, or wettable powder form.
- B. Pavement-Marking Paint: MPI #32 Alkyd Traffic Marking Paint.
 - 1. Color: As indicated on the plans.
- C. Pavement-Marking Paint: MPI #97 Latex Traffic Marking Paint.
 - 1. Color: As indicated on the plans.

- D. Glass Beads: AASHTO M 247, Type 1.
- E. Wheel Stops: Precast, air-entrained concrete, 2500-psi minimum compressive strength, 4-1/2 inches high by 9 inches wide by 72 inches long. Provide chamfered corners, drainage slots on underside, and holes for anchoring to substrate.
 - 1. Dowels: Galvanized steel, 3/4-inch diameter, 10-inch minimum length.
- F. Wheel Stops: Solid, integrally colored, 96 percent recycled HDPE or commingled postconsumer and postindustrial recycled plastic; UV stabilized; 4 inches high by 6 inches wide by 72 inches long. Provide chamfered corners, drainage slots on underside, and holes for anchoring to substrate.
 - 1. Dowels: Galvanized steel, 3/4-inch diameter, 10-inch minimum length.
 - 2. Adhesive: As recommended by wheel-stop manufacturer for application to asphalt pavement.

2.4 MIXES

- A. Hot-Mix Asphalt: Dense, hot-laid, hot-mix asphalt plant mixes approved by authorities having jurisdiction; designed according to procedures in AI MS-2, "Mix Design Methods for Asphalt Concrete and Other Hot-Mix Types"; and complying with the following requirements:
 - 1. Provide mixes with a history of satisfactory performance in geographical area where Project is located.
 - 2. Base Course: As indicated on the plans.
 - 3. Surface Course: As indicated on the plans.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Proof-roll subgrade below pavements with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.
- B. Proceed with paving only after unsatisfactory conditions have been corrected.

3.2 COLD MILLING

- A. Clean existing pavement surface of loose and deleterious material immediately before cold milling. Remove existing asphalt pavement by cold milling to grades and cross sections indicated.
 - 1. Mill to a depth of 2 inches.

3.3 PATCHING

- A. Hot-Mix Asphalt Pavement: Saw cut perimeter of patch and excavate existing pavement section to sound base. Excavate rectangular or trapezoidal patches, extending 12 inches into adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically. Remove excavated material. Recompact existing unbound-aggregate base course to form new subgrade.
- B. Portland Cement Concrete Pavement: Break cracked slabs and roll as required to reseal concrete pieces firmly.
 - 1. Remove disintegrated or badly cracked pavement. Excavate rectangular or trapezoidal patches, extending into adjacent sound pavement, unless otherwise indicated. Cut excavation faces vertically. Recompact existing unbound-aggregate base course to form new subgrade.
- C. Tack Coat: Apply uniformly to vertical surfaces abutting or projecting into new, hot-mix asphalt paving at a rate of 0.05 to 0.15 gal./sq. yd.
 - 1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
 - 2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.
- D. Patching: Fill excavated pavements with hot-mix asphalt base mix for full thickness of patch and, while still hot, compact flush with adjacent surface.

3.4 SURFACE PREPARATION

- A. General: Immediately before placing asphalt materials, remove loose and deleterious material from substrate surfaces. Ensure that prepared subgrade is ready to receive paving.
- B. Herbicide Treatment: Apply herbicide according to manufacturer's recommended rates and written application instructions. Apply to dry, prepared subgrade or surface of compacted-aggregate base before applying paving materials.
- C. Tack Coat: Apply uniformly to surfaces of existing pavement at a rate of 0.05 to 0.15 gal./sq. yd.
 - 1. Allow tack coat to cure undisturbed before applying hot-mix asphalt paving.
 - 2. Avoid smearing or staining adjoining surfaces, appurtenances, and surroundings. Remove spillages and clean affected surfaces.

3.5 HOT-MIX ASPHALT PLACING

- A. Machine place hot-mix asphalt on prepared surface, spread uniformly, and strike off. Place asphalt mix by hand to areas inaccessible to equipment in a manner that prevents segregation of mix. Place each course to required grade, cross section, and thickness when compacted.
 - 1. Spread mix at minimum temperature of 250 deg F.
 - 2. Regulate paver machine speed to obtain smooth, continuous surface free of pulls and tears in asphalt-paving mat.

- B. Place paving in consecutive strips not less than 10 feet wide unless infill edge strips of a lesser width are required.
- C. Promptly correct surface irregularities in paving course behind paver. Use suitable hand tools to remove excess material forming high spots. Fill depressions with hot-mix asphalt to prevent segregation of mix; use suitable hand tools to smooth surface.

3.6 JOINTS

- A. Construct joints to ensure a continuous bond between adjoining paving sections. Construct joints free of depressions, with same texture and smoothness as other sections of hot-mix asphalt course.
 - 1. Clean contact surfaces and apply tack coat to joints.
 - 2. Offset longitudinal joints, in successive courses, a minimum of 6 inches.
 - 3. Offset transverse joints, in successive courses, a minimum of 24 inches.
 - 4. Construct transverse joints at each point where paver ends a day's work and resumes work at a subsequent time. Construct these joints using either "bulkhead" or "papered" method according to AI MS-22, for both "Ending a Lane" and "Resumption of Paving Operations."

3.7 COMPACTION

- A. General: Begin compaction as soon as placed hot-mix paving will bear roller weight without excessive displacement. Compact hot-mix paving with hot, hand tampers or with vibratory-plate compactors in areas inaccessible to rollers.
 - 1. Complete compaction before mix temperature cools to 185 deg F.
- B. Breakdown Rolling: Complete breakdown or initial rolling immediately after rolling joints and outside edge. Examine surface immediately after breakdown rolling for indicated crown, grade, and smoothness. Correct laydown and rolling operations to comply with requirements.
- C. Intermediate Rolling: Begin intermediate rolling immediately after breakdown rolling while hot-mix asphalt is still hot enough to achieve specified density. Continue rolling until hot-mix asphalt course has been uniformly compacted to the following density:
 - 1. Average Density: 92 percent of reference maximum theoretical density according to ASTM D 2041, but not less than 90 percent nor greater than 96 percent.
- D. Finish Rolling: Finish roll paved surfaces to remove roller marks while hot-mix asphalt is still warm.
- E. Edge Shaping: While surface is being compacted and finished, trim edges of pavement to proper alignment. Bevel edges while asphalt is still hot; compact thoroughly.
- F. Protection: After final rolling, do not permit vehicular traffic on pavement until it has cooled and hardened.

- G. Erect barricades to protect paving from traffic until mixture has cooled enough not to become marked.

3.8 ASPHALT CURBS

- A. Construct hot-mix asphalt curbs over compacted pavement surfaces. Apply a light tack coat unless pavement surface is still tacky and free from dust. Spread mix at minimum temperature of 250 deg F.
 - 1. Asphalt Mix: Same as pavement surface-course mix.
- B. Place hot-mix asphalt to curb cross section indicated or, if not indicated, to local standard shapes, by machine or by hand in wood or metal forms. Tamp hand-placed materials and screed to smooth finish. Remove forms after hot-mix asphalt has cooled.

3.9 INSTALLATION TOLERANCES

- A. Pavement Thickness: Compact each course to produce the thickness indicated within the following tolerances:
 - 1. Base Course: Plus or minus 1/2 inch.
 - 2. Surface Course: Plus 1/4 inch, no minus.
- B. Pavement Surface Smoothness: Compact each course to produce a surface smoothness within the following tolerances as determined by using a 10-foot straightedge applied transversely or longitudinally to paved areas:
 - 1. Base Course: 1/4 inch.
 - 2. Surface Course: 1/8 inch.
 - 3. Crowned Surfaces: Test with crowned template centered and at right angle to crown. Maximum allowable variance from template is 1/4 inch.

3.10 PAVEMENT MARKING

- A. Do not apply pavement-marking paint until layout, colors, and placement have been verified with Architect.
- B. Allow paving to age for 30 days before starting pavement marking.
- C. Sweep and clean surface to eliminate loose material and dust.
- D. Apply paint with mechanical equipment to produce pavement markings, of dimensions indicated, with uniform, straight edges. Apply at manufacturer's recommended rates to provide a minimum wet film thickness of 15 mils.
 - 1. Broadcast glass beads uniformly into wet pavement markings at a rate of 6 lb/gal.

3.11 WHEEL STOPS

- A. Install wheel stops in bed of adhesive as recommended by manufacturer.
- B. Securely attach wheel stops to pavement with not less than two galvanized-steel dowels embedded at one-quarter to one-third points. Securely install dowels into pavement and bond to wheel stop. Recess head of dowel beneath top of wheel stop.

3.12 FIELD QUALITY CONTROL

- A. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.
- B. Replace and compact hot-mix asphalt where core tests were taken.
- C. Remove and replace or install additional hot-mix asphalt where test results or measurements indicate that it does not comply with specified requirements.

3.13 DISPOSAL

- A. Except for material indicated to be recycled, remove excavated materials from Project site and legally dispose of them in an EPA-approved landfill.

END OF SECTION

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SECTION 32 13 13
CONCRETE PAVING

PART 1 - GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Driveways.
 - 2. Roadways.
 - 3. Parking lots.
 - 4. Curbs and gutters.
 - 5. Walks.

1.2 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Samples: For each exposed product and for each color and texture specified.
- C. Other Action Submittals:
 - 1. Design Mixtures: For each concrete paving mixture. Include alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

1.3 QUALITY ASSURANCE

- A. Ready-Mix-Concrete Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.
- B. ACI Publications: Comply with ACI 301 unless otherwise indicated.

PART 2 - PRODUCTS

2.1 STEEL REINFORCEMENT

- A. Recycled Content: Provide steel reinforcement with an average recycled content of steel so postconsumer recycled content plus one-half of preconsumer recycled content is not less than 25 percent.

- B. Plain-Steel Welded Wire Reinforcement: ASTM A 185/A 185M, fabricated from as-drawn steel wire into flat sheets.
- C. Deformed-Steel Welded Wire Reinforcement: ASTM A 497/A 497M, flat sheet.
- D. Reinforcing Bars: ASTM A 615/A 615M, Grade 60; deformed.
- E. Plain-Steel Wire: ASTM A 82/A 82M, as drawn.
- F. Deformed-Steel Wire: ASTM A 496/A 496M.
- G. Dowel Bars: ASTM A 615/A 615M, Grade 60 plain-steel bars; zinc coated (galvanized) after fabrication according to ASTM A 767/A 767M, Class I coating. Cut bars true to length with ends square and free of burrs.
- H. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars, welded wire reinforcement, and dowels in place. Manufacture bar supports according to CRSI's "Manual of Standard Practice" from steel wire, plastic, or precast concrete of greater compressive strength than concrete specified.

2.2 CONCRETE MATERIALS

- A. Cementitious Material: Use the following cementitious materials, of same type, brand, and source throughout Project:
 - 1. Portland Cement: ASTM C 150, gray or white portland cement Type I, Type II, Type I/II, or Type V. Supplement with the following:
 - a. Fly Ash: ASTM C 618, Class F.
 - b. Ground Granulated Blast-Furnace Slag: ASTM C 989, Grade 100 or 120.
 - 2. Blended Hydraulic Cement: ASTM C 595, Type IP, portland-pozzolan cement.
- B. Normal-Weight Aggregates: ASTM C 33, Class 4S, Class 4M, or Class 1N, uniformly graded. Provide aggregates from a single source.
- C. Water: Potable and complying with ASTM C 94/C 94M.
- D. Air-Entraining Admixture: ASTM C 260.
- E. Chemical Admixtures: Admixtures certified by manufacturer to be compatible with other admixtures and to contain no more than 0.1 percent water-soluble chloride ions by mass of cementitious material.
- F. Color Pigment: ASTM C 979, synthetic mineral-oxide pigments or colored water-reducing admixtures; color stable, free of carbon black, nonfading, and resistant to lime and other alkalis.
 - 1. Color: As selected by Architect from manufacturer's full range.

2.3 FIBER REINFORCEMENT

- A. Synthetic Fiber: Monofilament or fibrillated polypropylene fibers engineered and designed for use in concrete paving, complying with ASTM C 1116/C 1116M, Type III, 1/2 to 1-1/2 inches long.

2.4 CURING MATERIALS

- A. Absorptive Cover: AASHTO M 182, Class 3, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. dry or cotton mats.
- B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.
- C. Water: Potable.
- D. Evaporation Retarder: Waterborne, monomolecular, film forming, manufactured for application to fresh concrete.
- E. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, dissipating.
- F. White, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 2, Class B, dissipating.

2.5 RELATED MATERIALS

- A. Joint Fillers: ASTM D 1751, asphalt-saturated cellulosic fiber or ASTM D 1752, cork or self-expanding cork in preformed strips.
- B. Slip-Resistive Aggregate Finish: Factory-graded, packaged, rustproof, nonglazing, abrasive aggregate of fused aluminum-oxide granules or crushed emery aggregate containing not less than 50 percent aluminum oxide and not less than 20 percent ferric oxide; unaffected by freezing, moisture, and cleaning materials.

2.6 PAVEMENT MARKINGS

- A. Pavement-Marking Paint: Latex, waterborne emulsion, lead and chromate free, ready mixed, complying with FS TT-P-1952, Type II, with drying time of less than 45 minutes.
 - 1. Color: As indicated on plans.
- B. Pavement-Marking Paint: MPI #97 Latex Traffic Marking Paint.
 - 1. Color: As indicated on plans.

2.7 WHEEL STOPS

- A. Wheel Stops: Precast, air-entrained concrete or Solid, integrally colored, 96 percent recycled HDPE, or commingled postconsumer and postindustrial recycled rubber or plastic; UV stabilized.
 - 1. Color: As indicated on plans.
 - 2. Dowels: Galvanized steel, 3/4 inch in diameter, 10-inch minimum length.
 - 3. Adhesive: As recommended by wheel stop manufacturer for application to concrete pavement.

2.8 CONCRETE MIXTURES

- A. Prepare design mixtures, proportioned according to ACI 301, with the following properties:
 - 1. Compressive Strength (28 Days): As indicated on plans.
 - 2. Maximum Water-Cementitious Materials Ratio at Point of Placement: 0.45.
 - 3. Slump Limit: 4 inches to 8 inches, plus or minus 1 inch.
 - 4. Air Content: 4-1/2 to 6 percent plus or minus 1.5 percent.
- B. Chemical Admixtures: Use admixtures according to manufacturer's written instructions.
- C. Synthetic Fiber: Uniformly disperse in concrete mixture at manufacturer's recommended rate, but not less than 1.0 lb/cu. yd.
- D. Color Pigment: Add color pigment to concrete mixture according to manufacturer's written instructions.

2.9 CONCRETE MIXING

- A. Ready-Mixed Concrete: Measure, batch, and mix concrete materials and concrete according to ASTM C 94/C 94M and ASTM C 1116/C 1116M. Furnish batch certificates for each batch discharged and used in the Work.

PART 3 - EXECUTION

3.1 EXAMINATION AND PREPARATION

- A. Proof-roll prepared subbase surface below concrete paving to identify soft pockets and areas of excess yielding.
- B. Remove loose material from compacted subbase surface immediately before placing concrete.

3.2 EDGE FORMS AND SCREED CONSTRUCTION

- A. Set, brace, and secure edge forms, bulkheads, and intermediate screed guides to required lines, grades, and elevations. Install forms to allow continuous progress of work and so forms can remain in place at least 24 hours after concrete placement.
- B. Clean forms after each use and coat with form-release agent to ensure separation from concrete without damage.

3.3 STEEL REINFORCEMENT

- A. General: Comply with CRSI's "Manual of Standard Practice" for fabricating, placing, and supporting reinforcement.

3.4 JOINTS

- A. General: Form construction, isolation, and contraction joints and tool edges true to line, with faces perpendicular to surface plane of concrete. Construct transverse joints at right angles to centerline unless otherwise indicated.
- B. Construction Joints: Set construction joints at side and end terminations of paving and at locations where paving operations are stopped for more than one-half hour unless paving terminates at isolation joints.
- C. Isolation Joints: Form isolation joints of preformed joint-filler strips abutting concrete curbs, catch basins, manholes, inlets, structures, other fixed objects, and where indicated.
- D. Contraction Joints: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of the concrete thickness or to match jointing of existing adjacent concrete paving.
- E. Edging: After initial floating, tool edges of paving, gutters, curbs, and joints in concrete with an edging tool to a 1/4-inch radius. Repeat tooling of edges after applying surface finishes. Eliminate edging-tool marks on concrete surfaces.

3.5 CONCRETE PLACEMENT

- A. Moisten subbase to provide a uniform dampened condition at time concrete is placed.
- B. Comply with ACI 301 requirements for measuring, mixing, transporting, placing, and consolidating concrete.
- C. Deposit and spread concrete in a continuous operation between transverse joints. Do not push or drag concrete into place or use vibrators to move concrete into place.
- D. Screed paving surface with a straightedge and strike off.

- E. Commence initial floating using bull floats or darbies to impart an open-textured and uniform surface plane before excess moisture or bleed water appears on the surface. Do not further disturb concrete surfaces before beginning finishing operations or spreading surface treatments.

3.6 FLOAT FINISHING

- A. General: Do not add water to concrete surfaces during finishing operations.
- B. Float Finish: Begin the second floating operation when bleed-water sheen has disappeared and concrete surface has stiffened sufficiently to permit operations. Float surface with power-driven floats or by hand floating if area is small or inaccessible to power units. Finish surfaces to true planes. Cut down high spots and fill low spots. Refloat surface immediately to uniform granular texture.
 - 1. Burlap Finish: Drag a seamless strip of damp burlap across float-finished concrete, perpendicular to line of traffic, to provide a uniform, gritty texture.
 - 2. Medium-to-Fine-Textured Broom Finish: Draw a soft-bristle broom across float-finished concrete surface perpendicular to line of traffic to provide a uniform, fine-line texture.
 - 3. Medium-to-Coarse-Textured Broom Finish: Provide a coarse finish by striating float-finished concrete surface 1/16 to 1/8 inch deep with a stiff-bristled broom, perpendicular to line of traffic.
- C. Slip-Resistive Aggregate Finish: Before final floating, spread slip-resistive aggregate finish on paving surface according to manufacturer's written instructions.
 - 1. Cure concrete with curing compound recommended by slip-resistive aggregate manufacturer. Apply curing compound immediately after final finishing.
 - 2. After curing, lightly work surface with a steel wire brush or abrasive stone and water to expose nonslip aggregate.

3.7 CONCRETE PROTECTION AND CURING

- A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures.
- B. Comply with ACI 306.1 for cold-weather protection.
- C. Evaporation Retarder: Apply evaporation retarder to concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete but before float finishing.
- D. Begin curing after finishing concrete but not before free water has disappeared from concrete surface.
- E. Curing Methods: Cure concrete by moisture curing, moisture-retaining-cover curing or curing compound or a combination of these.

3.8 PAVING TOLERANCES

- A. Comply with tolerances in ACI 117 and as follows:
 - 1. Elevation: 3/4 inch.
 - 2. Thickness: Plus 3/8 inch, minus 1/4 inch.
 - 3. Surface: Gap below 10-foot-long, unleveled straightedge not to exceed 1/2 inch.
 - 4. Joint Spacing: 3 inches.
 - 5. Contraction Joint Depth: Plus 1/4 inch, no minus.
 - 6. Joint Width: Plus 1/8 inch, no minus.

3.9 PAVEMENT MARKING

- A. Allow concrete paving to cure for a minimum of 28 days and be dry before starting pavement marking.
- B. Sweep and clean surface to eliminate loose material and dust.
- C. Apply paint with mechanical equipment to produce markings of dimensions indicated with uniform, straight edges. Apply at manufacturer's recommended rates to provide a minimum wet film thickness of 15 mils.

3.10 WHEEL STOPS

- A. Install wheel stops in bed of adhesive applied as recommended by manufacturer.
- B. Securely attach wheel stops to paving with not less than two steel dowels located at one-quarter to one-third points. Install dowels in drilled holes in the paving and bond dowels to wheel stop. Recess head of dowel beneath top of wheel stop.

3.11 REPAIRS AND PROTECTION

- A. Remove and replace concrete paving that is broken, damaged, or defective or that does not comply with requirements in this Section. Remove work in complete sections from joint to joint unless otherwise approved by Architect.
- B. Protect concrete paving from damage. Exclude traffic from paving for at least 14 days after placement. When construction traffic is permitted, maintain paving as clean as possible by removing surface stains and spillage of materials as they occur.
- C. Maintain concrete paving free of stains, discoloration, dirt, and other foreign material. Sweep paving not more than two days before date scheduled for Substantial Completion inspections.

END OF SECTION

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SECTION 32 13 73

CONCRETE PAVING JOINT SEALANTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Cold-applied joint sealants.
2. Cold-applied, jet-fuel-resistant joint sealants.
3. Hot-applied joint sealants.
4. Hot-applied, jet-fuel-resistant joint sealants.

1.2 PRECONSTRUCTION TESTING

- ###### A. Preconstruction Compatibility and Adhesion Testing:
- Submit to joint-sealant manufacturers, eight Samples of materials that will contact or affect joint sealants. Use ASTM C 1087 to determine whether priming and other specific joint-preparation techniques are required to obtain rapid, optimum adhesion of joint sealants to joint substrates.

1.3 SUBMITTALS

- ###### A. Product Data:
- For each joint-sealant product indicated.
- ###### B. Samples:
- For each kind and color of joint sealant required.
- ###### C. Pavement-Joint-Sealant Schedule:
- Include the following information:
1. Joint-sealant application, joint location, and designation.
 2. Joint-sealant manufacturer and product name.
 3. Joint-sealant formulation.
 4. Joint-sealant color.
- ###### D. Product certificates.
- ###### E. Product test reports.
- ###### F. Preconstruction compatibility and adhesion test reports.

1.4 QUALITY ASSURANCE

- ###### A. Testing Agency Qualifications:
- An independent testing agency qualified according to ASTM C 1021.

- B. Preinstallation Conference: Conduct conference at Tonopah Elementary School.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Compatibility: Provide joint sealants, backing materials, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by joint-sealant manufacturer based on testing and field experience.
- B. Colors of Exposed Joint Sealants: As selected by Architect from manufacturer's full range.

2.2 COLD-APPLIED JOINT SEALANTS

- A. Single-Component, Nonsag, Silicone Joint Sealant for Concrete: ASTM D 5893, Type NS.
 - 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Crafcro Inc., an ERGON company; RoadSaver Silicone.
 - b. Dow Corning Corporation; 888.
 - c. Pecora Corporation; 301 NS.
- B. Single-Component, Self-Leveling, Silicone Joint Sealant for Concrete: ASTM D 5893, Type SL.
 - 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Crafcro Inc., an ERGON company; RoadSaver Silicone SL.
 - b. Dow Corning Corporation; 890-SL.
 - c. Pecora Corporation; 300 SL.
- C. Multicomponent, Pourable, Traffic-Grade, Urethane Joint Sealant for Concrete: ASTM C 920, Type M, Grade P, Class 25, for Use T.
 - 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Pecora Corporation; Urexpan NR-200.

2.3 COLD-APPLIED, JET-FUEL-RESISTANT JOINT SEALANTS

- A. Jet-Fuel-Resistant, Single-Component, Pourable, Traffic-Grade, Modified-Urethane Joint Sealant for Concrete: ASTM C 920, Type S, Grade P, Class 25, for Use T.

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. BASF Building Systems; Sonomeric 1.
- B. Jet-Fuel-Resistant, Multicomponent, Pourable, Traffic-Grade, Modified-Urethane Joint Sealant for Concrete: ASTM C 920, Type M, Grade P, Class 12-1/2, for Use T.
 1. Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Pecora Corporation; Urexpan NR-300.
- C. Jet-Fuel-Resistant, Multicomponent, Pourable, Traffic-Grade, Modified-Urethane Joint Sealant for Concrete: ASTM C 920, Type M, Grade P, Class 25, for Use T.
 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Meadows, W. R., Inc.; Sealtight Gardox.

2.4 HOT-APPLIED JOINT SEALANTS

- A. Hot-Applied, Single-Component Joint Sealant for Concrete: ASTM D 3406.
 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Crafcro Inc., an ERGON company; Superseal 444/777.
- B. Hot-Applied, Single-Component Joint Sealant for Concrete and Asphalt: ASTM D 6690, Types I, II, and III.
 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Meadows, W. R., Inc.; Sealtight Hi-Spec or Sealtight 3405.
 - b. Right Pointe; D-3405 Hot Applied Sealant.

2.5 HOT-APPLIED, JET-FUEL-RESISTANT JOINT SEALANTS

- A. Hot-Applied, Jet-Fuel-Resistant, Single-Component Joint Sealant for Concrete: ASTM D 7116, Type I.
 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Crafcro Inc., an ERGON company; Superseal 444/777.

- B. Hot-Applied, Jet-Fuel-Resistant, Single-Component Joint Sealant for Concrete and Tar Concrete: Single-component formulation complying with ASTM D 3581.
 - 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Crafcro Inc., an ERGON company; Superseal 1614A.

2.6 JOINT-SEALANT BACKER MATERIALS

- A. Round Backer Rods for Cold- and Hot-Applied Joint Sealants: ASTM D 5249, Type 1, of diameter and density required to control sealant depth and prevent bottom-side adhesion of sealant.
- B. Round Backer Rods for Cold-Applied Joint Sealants: ASTM D 5249, Type 3, of diameter and density required to control joint-sealant depth and prevent bottom-side adhesion of sealant.
- C. Backer Strips for Cold- and Hot-Applied Joint Sealants: ASTM D 5249; Type 2; of thickness and width required to control joint-sealant depth, prevent bottom-side adhesion of sealant, and fill remainder of joint opening under sealant.

2.7 PRIMERS

- A. Primers: Product recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Comply with joint-sealant manufacturer's written installation instructions for products and applications indicated unless more stringent requirements apply.
- B. Cleaning of Joints: Clean out joints immediately before installing joint sealants.
- C. Joint-Sealant Installation Standard: Comply with recommendations in ASTM C 1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.
- D. Install joint-sealant backings of kind indicated to support joint sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.
 - 1. Do not leave gaps between ends of joint-sealant backings.
 - 2. Do not stretch, twist, puncture, or tear joint-sealant backings.
 - 3. Remove absorbent joint-sealant backings that have become wet before sealant application and replace them with dry materials.

- E. Install joint sealants using proven techniques that comply with the following and at the same time backings are installed:
 - 1. Place joint sealants so they directly contact and fully wet joint substrates.
 - 2. Completely fill recesses in each joint configuration.
 - 3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.
- F. Tooling of Nonsag Joint Sealants: Immediately after joint-sealant application and before skinning or curing begins, tool sealants according to the following requirements to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint:
 - 1. Remove excess joint sealant from surfaces adjacent to joints.
 - 2. Use tooling agents that are approved in writing by joint-sealant manufacturer and that do not discolor sealants or adjacent surfaces.
- G. Provide joint configuration to comply with joint-sealant manufacturer's written instructions unless otherwise indicated.
- H. Clean off excess joint sealant or sealant smears adjacent to joints as the Work progresses, by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

END OF SECTION

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SECTION 32 1623 SIDEWALKS

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Concrete sidewalks.
- B. Concrete wheelchair ramps.

1.02 RELATED REQUIREMENTS

- A. Section 32 1123 - Aggregate Base Courses.
- B. Section 32 1313 - Concrete Paving.

1.03 REFERENCE STANDARDS

- A. ACI PRC-211.1 - Selecting Proportions for Normal-Density and High Density-Concrete - Guide; 2022.
- B. ACI PRC-305 - Guide to Hot Weather Concreting; 2020.
- C. ACI PRC-306 - Guide to Cold Weather Concreting; 2016.
- D. ADA Standards - 2010 ADA Standards for Accessible Design; 2010.
- E. ASTM C33/C33M - Standard Specification for Concrete Aggregates; 2023.
- F. ASTM C94/C94M - Standard Specification for Ready-Mixed Concrete; 2023.

1.04 SUBMITTALS

- A. See Section 01 3000 - Administrative Requirements for submittal procedures.

1.05 FIELD CONDITIONS

- A. Follow recommendations of ACI PRC-305 and ACI PRC-306 when concreting during hot and cold weather, respectively.

PART 2 PRODUCTS

2.01 CONCRETE SIDEWALKS AND WHEELCHAIR RAMPS

- A. Concrete Forms: Wood.
- B. Concrete Materials: Comply with ASTM C94/C94M.
- C. Aggregate: Pit Run, washed, 3/8 inch (1 cm) stone; free of shale, clay, friable material and debris.
- D. Joint Filler: Preformed expansion, with a thickness of 1/2 inch (13 mm).

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify gradients and elevations of the subgrade are correct as shown on drawings. Where poor subgrade material is encountered, remove and replace with suitable material.
- B. Verify compacted subgrade is acceptable, ready to support imposed loads and paving, and ready to receive work.

3.02 SUBBASE PREPARATION

- A. Maintain subgrade in a smooth, compacted condition with required section and established grade until concrete is placed.
- B. See Section 32 1123 for aggregate subbase.

3.03 CONCRETE SIDEWALK AND WHEELCHAIR RAMP INSTALLATION

- A. Forming:
 - 1. Assemble formwork to permit easy stripping and dismantling without damaging concrete.

2. Sidewalk Forms: Place and secure forms to location, dimension, profile, and gradient shown on drawings. Height equal to the full depth of the finished sidewalk.
 3. Wheelchair Ramps: Place and secure forms to location, dimension, profile, and gradient shown on drawings. Comply with ADA Standards.
- B. Placement:
1. Place concrete in a single lift.
 2. Consolidate concrete by tamping and spading.
- C. Joints:
1. Spacing: Provide scored joints every 10 feet (3 m).
 2. Layout: See site plan drawing for proposed joint locations.
 3. Filler height equal to the full depth of the finished concrete.
- D. Finishing:
1. Sidewalk Paving: Light broom, texture perpendicular to direction of travel with troweled and radiused edge, 1/4 inch radius (6 mm radius).
 2. Wheelchair Ramps: Broomed perpendicular to slope.

3.04 TOLERANCES

- A. Surface Flatness: 1/4 inch (6 mm), maximum, measured with 10 foot (3 m) straight edge.

END OF SECTION

SECTION 32 18 13

SYNTHETIC TURF

PART 1 GENERAL

1.1 WORK INCLUDED

- A. Furnish all labor, materials, tools, equipment, and shop drawings and calculations necessary to install all synthetic turf as indicated on the plans and as specified herein. The installation of all new materials shall be performed in strict accordance with the manufacturer's installation instructions and in accordance with all approved shop drawings.
- B. Perimeter edge details required for the synthetic turf system and drainage details (as necessary) shall be as detailed and recommended by the Manufacturer and shall be approved by the A/E and Owner's Representative.
- C. Work shall include all components and accessories required for a complete installation, including but not limited to
 - 1. Acceptance of prepared sub-base.
 - 2. Coordination with related trades to ensure a complete, integrated, and timely installation: Aggregate base course, sub-base material (tested for permeability), grading and compacting, piping and drain components (when required); as provided under its respective trade section.

1.2 RELATED SECTIONS

- A. Section 31 1100 – Clearing and Grubbing
- B. Section 31 2300 – Excavation and Fill.
- C. Section 32 84 00 – Landscape Irrigation

1.3 REFERENCES

- A. FM P7825 - Approval Guide; Factory Mutual Research Corporation; current edition
- B. ASTM Standard Test Methods:
 - D1577 Standard Test Method for Linear Density of Textile Fiber
 - D5848 Standard Test Method for Mass Per Unit Area of Pile Yarn Floor Covering
 - D418 Standard Test Method for Testing Pile Yarn Floor Covering Construction
 - D1335 Standard Test Method for Tuft Bind of Pile Yarn Floor Coverings
 - D1682 Standard Method of Test for Breaking Load and Elongation of Textile Fabrics
 - D5034 Standard Test Method of Breaking Strength and Elongation of Textile Fabrics (Grab Test)
 - F1015 Standard Test Method for Relative Abrasiveness of Synthetic Turf Playing Surfaces
 - F2117 Standard Test Method for Vertical Rebound Characteristics of Sports surface/Ball Systems.
 - F1551 Standard Test Methods for Comprehensive Characterization of Synthetic Turf Playing Surfaces and Materials
 - D4491 Standard Test Methods for Water Permeability of Geotextiles by Permittivity
 - D2859 Standard Test Method for Ignition Characteristics of Finished Textile Floor Covering Materials.
 - F355 Standard Test Method for Shock-Absorbing Properties of Playing Surfaces.

- F1936 Standard Test Method for Shock-Absorbing Properties of North American Football Field Playing Systems as Measured in the Field.
- D1557 Test Method for Laboratory Compaction Characteristics of Soil Using Modified Effort.
- E-11 Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
- E-648 Flooring Critical Radiant Panel Test (Flame Spread)

1.4 SUBMITTALS

- A. Product data: the Contractor shall submit complete material submittals, including manufacturer's name and address, specific trade names, catalog and model numbers, samples, warranty information, illustrations and descriptive material, clearly marked as to proposed items for approval by the A/E and/or Owner's Representative.
- B. Comply with section 01 33 00, Submittals Procedures. Submit product data for approval prior to fabrication.
- C. The following list of submittals is a requirement of this specification and must be provided for review before any approval will be given.
 - 1. Shop Drawings. Complete and detailed shop drawings including layout of all components, parts, and materials necessary for a complete synthetic turf system. Shop drawings shall show proposed locations of all seams in fabric surfacing.
 - 2. Submit a complete scale and dimensioned drawing(s) of inlaid or tufted-in field lining and marking boundaries.
 - 3. Submit two samples, 12"x12" in size, illustrating of actual turf surface; provide 1 sample of turf surface material with infill in standard sample container.
 - 4. One sample each, 12" long sample of 4" white line and other colored lines, without infill.
 - 5. Provide one 12"x12" sample of the specified seaming method.
 - 6. Provide one 12"x12" sample of the specified pad system.
 - 7. Minimum 1/2 pound of the proposed infill mix.
 - 8. A letter and specification sheet certifying that the products of this section meet or exceed specified requirements and that the turf system products meet or exceed the testing standards below.
 - a. Certified copies of independent (third-party) laboratory reports on ASTM tests as follows:

1) ASTM F1015-03	Relative abrasive index
2) ASTM F2117-01	Average ball rebound height (Entire System: turf, infill & pad)
3) ASTM F1551-03	Coefficient of restitution (CR)
4) ASTM F1551-03	Average ball bounce (Entire System: turf, infill & pad)
5) ASTM F1551-03	Permeability of Turf
6) ASTM F1551-03	Soccer shoe traction WET and DRY (Entire System: turf, infill & pad)
7) ASTM F1551-03	Football shoe traction WET and DRY (Entire System: turf, infill & pad)
8) ASTM D5848-05	Total Weight
9) ASTM D5848-05	Pile Weight
10) ASTM D5848-05	Primary Backing Weight
11) ASTM D5848-05	Secondary Backing
12) ASTM D5823-05a	Pile Height
13) ASTM D1335-05	Tuft bind strength
14) ASTM D5034-95R01	Grab tear strength
15) ASTM D5793-05	Stitches per 3 inches
16) ASTM D5793-05	Machine gauge
17) ASTM D2859-04	Flammability – Pill burn
18) ASTM F1951-99	Wheelchair accessibility
19) BS7044-Method 4	Infiltration Rate
20) ASTM D1907-07	Fiber denier
21) ASTM D3218-07	Fiber thickness

- 22) ASTM D3218-07 Fiber width
- 23) ASTM D789-07 Fiber melting point
- 24) ASTM D0792-00 Fiber & infill specific gravity
- 25) ASTM D2256-02 Fiber elongation
- 26) ASTM E0648-04 Flooring critical radiant panel test (flame spread)
- 27) ASTM F0355-01 Shock absorbing properties of pad

- b. ASTM F1936-98 Shock-Absorbing Properties of North American Football Field Playing Systems as Measured in the Field
- c. ASTM D3218-01 Average thickness and resin classification.
- d. ASTM D4716-04 Hydraulic Transmittivity (pad tested at 0.5% slope under 2pka stress).
- 9. Certified copies of independent (third-party) laboratory reports on ASTM tests for harmful metals as follows:
 - a. ASTM 3052@210 °Lead- Submit a lead and heavy metal letter, outlining that the synthetic turf system proposed is in compliance with the proposed 2011 standard of a maximum of 100 parts per million for the entire synthetic turf system including infill for lead and heavy metals. No nylon fibers are allowed.
 - b. RCRA 7471B Mercury (HG)
 - c. RCRA 6010`Metals
- 10. Lists providing specific contacts and telephone numbers of eight (8) other projects of similar size and scope, using the same product.
- 11. Résumé of Installation Supervisor who will be present on-site during installation.
- 12. Prior to the A/E and/or Owner's Representative's approval of a specified synthetic turf system, the Manufacturer shall specify in writing that their turf system does not violate any other manufacturer's patents, patents allowed or patents pending.
- D. Approval of the submittals shall be the Contractor's authorization to order the required material. There will be no deviation from the approved submittals without the written authorization of the A/E or Owner's Representative.
- E. Prior to ordering of materials:
 - 1. The Contractor shall submit Shop Drawings indicating:
 - a. Field Layout
 - b. Field Marking Plan and details for the specified sports (if required)
 - c. Roll/Seaming Layout
 - d. Methods of attachment, field openings and perimeter conditions.
 - e. Sub-base requirements including aggregate locations, sizes and depths; and liner locations.
 - f. Padding and under Drain System shop drawings and specifications, including layout and pipe sizes.
 - 2. The turf Manufacturer must submit the fiber manufacturer's name, type of fiber and composition of fiber.
- F. Requirements to be tested after installation of football or soccer fields:
 - 1. Contractor to provide independent 3rd party testing by a laboratory specializing in sports surfaces field testing.
 - 2. Contractor to provide documentation showing turf system meets Head Injury Coefficient (HIC) and G-Max values as required and listed in this specification.
 - 3. Required HIC Test Method
 - a. The required test method for determining the various Head Injury Coefficient (HIC) values as detailed within this section is ASTM F355-E – Standard Test Method for Shock-Absorbing Properties of Playing Surface Systems and Materials and ASTM F1936- Standard Test Method for Shock-Absorbing Properties of North American Football Field Playing Systems as Measured in the Field.
 - 4. Required HIC Values and Limits
 - a. At time of installation:
 - 1) The maximum HIC Value at time of installation shall not exceed 900 at 1.3 meter

- drop height.
 - 2) The HIC Value throughout the warranted lifecycle of the synthetic turf playing surface is not to exceed 1000 at 1.3 meter drop height.
 - 5. Required G-Max Test Method
 - a. The required test method for determining the various G-Max values as detailed within this section is ASTM F355-E – Standard Test Method for Shock-Absorbing Properties of Playing Surface Systems and Materials and ASTM F1936- Standard Test Method for Shock-Absorbing Properties of North American Football Field Playing Systems as Measured in the Field.
 - 6. Required G-Max Values and Limits
 - a. At time of installation:
 - 1) The maximum G-Max Value at time of installation shall not exceed 120.
 - 2) The G-Max Value throughout the warranted lifecycle of the synthetic turf playing surface is not to exceed 165.
- G. Prior to Final Acceptance, the Contractor shall submit to the Owner:
 - 1. Three (3) copies of Maintenance Manuals, which will include all necessary instructions for the proper care and preventative maintenance of the synthetic turf system, including, but not limited to:
 - a. General Maintenance
 - b. Lines and Mark Removal – Paint
 - c. Small Repair Procedures
 - d. Cleaning
 - e. A list of all procedures to avoid in order to prolong the life and maintain the warranty, including static and dynamic load limits, snow clearing, etc.
 - 2. Project Record Documents: Record actual locations of seams, drains or other pertinent information.
 - 3. Warranty: Submit Manufacturer Warranty and ensure that forms have been completed in Owner's name and registered with Manufacturer.
 - 4. Warranty Insurance Policy Certificate.
- H. Closeout Documents, the Contractor shall submit as part of the Operation and Maintenance Documents:
 - 1. Product Data Sheet for the proposed turf product with relative characteristics and all associated details.
 - 2. All testing documents for the actual product, conducted by a third-party laboratory, as required and listed in the specifications.
 - 3. Submit a copy of the Turf Manufacturers maintenance manual.
 - 4. Submit a copy of the Turf Manufacturer installation standards.
 - 5. Submit a copy of the Turf Manufacturer environmental and recycling policy.
 - 6. Copies of the Warranty and Guarantee requirements required and listed in this specification.
 - 7. Copies of drawing submittals and as-builts required and listed in this specification.

1.5 QUALITY ASSURANCE

- A. Comply with section 01 33 00, Submittal Procedures
- B. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section. The Turf Contractor and/or the turf Manufacturer:
 - 1. Must be experienced in the manufacture and installation of this specific type of synthetic infill grass system with the same manufacturer, product and company they are proposing for this field. This includes the same pile fiber, the backing, the backing coating, and the installation method.
- C. Installer Qualifications: Company specializing in performing the work of this section with minimum 5 years of experience installing synthetic turf systems on sports fields.

1. The Turf Contractor must provide competent workmen skilled in this specific type of synthetic grass installation.
 2. The designated Supervisory Personnel on the project must be certified, in writing by the turf Manufacturer, as competent in the installation of this material, including sewing or gluing seams and proper installation of the infill mixture.
 3. The Manufacturer shall have a Representative on site to certify the installation and Warranty compliance.
- D. Prior to the beginning of installation, the Installer of the synthetic turf, the Authority Having Jurisdiction, and/or the A/E shall inspect the sub-base and site work. The installer will accept the sub-base in writing when test results for compaction, planarity and permeability that are in compliance with the synthetic turf manufacturer's guidelines are provided. The Installer shall have the dimensions of the field and locations for markings measured by a registered surveyor to verify conformity to the specifications and applicable standards. A record of the finished field as-built measurements shall be made. The final tolerance-to-grade of the base shall not exceed ¼ inch in ten feet in any direction.
- E. The Turf Contractor shall provide the necessary testing data to the Authority Having Jurisdiction, and/or A/E that the finished field meets the required shock attenuation, as per ASTM F1936. The maximum acceptable HIC value at installation will be less than 800 for football and soccer fields.

1.6 ONLINE PRE-INSTALLATION MEETING

- A. Convene minimum of one week before starting work of this section.

1.7 DELIVERY, STORAGE, AND PROTECTION

- A. Deliver products to project site in wrapped condition.
- B. Store products under cover and elevated above grade.

1.8 WARRANTY

- A. See Section 01 77 00 - Closeout Procedures, for additional warranty requirements.
- B. The turf Manufacturer shall provide a Warranty to the Owner that covers defects in materials and workmanship of the turf for a minimum period of eight (8) years from the date of Substantial Completion. The turf manufacturer must verify that their onsite Representative has inspected the installation and that the work conforms to the manufacturer's requirements.
- C. The Manufacturer's Warranty shall include general wear and damage caused from UV degradation. The warranty shall specifically exclude vandalism, and acts of God beyond the control of the Owner or the manufacturer.
- D. The turf Manufacturer's Warranty must be supported by an insurance policy with an A rated insurance company for the full eight (8) year period. The policy cannot be a re-insurance or offshore policy. Coverage and exclusions must be detailed as part of the Bid Submittal process, including the following as a minimum:
1. Length of policy
 2. Name of the carrier
 3. Method for payment of the policy (must be paid up front and policy must be non-cancelable)
 4. Limit for single claims
 5. Limit for aggregate claims
 6. Additional policy features of importance

- E. The Turf Contractor shall provide a Warranty to the Owner that covers defects in the installation workmanship, and further warrant that the installation was done in accordance with both the Manufacturer's recommendations and any written directives of the Manufacturer's onsite Representative.
- F. The pad Manufacturer shall provide a Warranty to the Owner that covers defects in materials and workmanship of the pad for a minimum period of twenty (20) years from the date of Substantial Completion. The pad manufacturer must verify that their onsite Representative has inspected the installation and that the work conforms to the manufacturer's requirements.
- G. The synthetic turf system must maintain an ASTM 355 HIC less than 1000 for the life of the turf Warranty.

1.9 PROJECT RECORD DOCUMENTS

- A. Record actual locations of seams, drains and other pertinent information in accordance with Division 1 Specifications Series, General Requirements.

1.10 MAINTENANCE SERVICE

- A. The Turf Contractor will train the Owner's facility maintenance staff and others in the use of the turf Manufacturer's recommended groomer.

1.11 EXTRA MATERIALS

- A. Upon completion of work Contractor shall provide the following extra materials to the Owner:
 - 1. All fields:
 - a. 150 square feet of synthetic turf material for each color and height combination, for the soccer and football field.
 - b. 150 square feet of pad material for each type of pad used.
 - c. 500 pounds of turf infill material.

PART 2 PRODUCTS

- 2.1 Synthetic Turf System Products. See drawings for type, configuration, and colors of synthetic turf to be provided and installed. If all of the parameters, specifications and design intent of the drawings are met, the following list of manufacturers with an approved product would be acceptable for use upon approval: "approved equal" substitutions may be acceptable but must be submitted under the provisions of Section 01 25 00.

Astro Turf
706-277-8873

Hellas
512-250-2910

Shaw Sports Turf
657-557-5571

SporTurf
800-798-1056

Valley Precision Grading
916-638-8800

SprinTurf
530-249-0581

ACT Global
512-733-5300

2.2 MATERIALS FOR FOOTBALL & SOCCER FIELDS

- A. The component materials of the synthetic turf system consist of:
1. Fabric Surface: Made of parallel slit film or monofilament polyethylene fibers, helix or straight shaped, or approved fiber. Fibers shall be tufted to a woven polypropylene primary backing then coated with a secondary polyurethane synthetic coating material. The fiber shall be non-abrasive, proven athletic caliber, designed specifically for outdoor use and treated with UV inhibitor and stabilizers to resist the effects of ultraviolet degradation, heat, foot traffic, water and airborne pollutants. The fiber shall contain no toxic substances or heavy metals. The entire system shall be resistant to weather, insects, rot, mildew and fungus growth and will be non-allergic and non-toxic.
 - a. Pile Height Football/Soccer Turf- Minimum 1.625" pile height. (Pile height to accommodate selected infill material.)
 - b. Fiber Thickness- 100 Microns minimum.
 - c. Linear Density (Denier)- 9,000 minimum.
 - d. Average Pile Yarn Weight- 46 oz/square yard minimum.
 - e. Primary Backing Weight- 7.5 oz/square yard minimum.
 - f. Secondary Backing Weight- 20 oz/square yard minimum.
 - g. Turf Bind Strength- 8.0 lbs/force minimum.
 - h. Fiber Elongation- 45% minimum.
 - i. Fiber Specific Gravity- 0.90 minimum.
 - j. Fiber Melting Point- 220 degrees Fahrenheit/104 Celcius.
 - k. Permeability- ten (10) inches per hour minimum.
 - l. Roll Width- 15'.
 - m. Despite the inlaid lines and logos, the pile surface shall be suitable for both temporary and permanent line markings using acrylic paint, as per the manufacturer's specifications.
 - n. All adhesives used in bonding the seams and inlaid markings to the supplemental backing material shall be resistant to moisture, bacterial and fungal attacks and resistant to UV radiation.
 - o. The pile surface shall be nominally uniform in length.
 - p. Synthetic turf must have been tested for a minimum of 20,000 cycles in the Lisport Test machine. Report to be supplied by Contractor.
 2. Infill Material:
 - a. No natural, recycled, SBR, or EPDM rubber materials.
 - b. An infill installed per manufacturer's specifications and instructions consisting of one of the following:
 - 1) Engineered wood particle (BrockFill by Brock International) at 1.17 lbs per square foot over raw round sand at 4.5 lbs per square foot.
 - 2) Ecotherm (by Polygreen) at 2 lbs per square foot over RealFill (by Hellas) at 3 lbs per square foot.
 - 3) Acrylic coated round sand infused with anti-microbial technology at 6 lbs per square foot minimum.
 - 4) Walnut shells (Safeshell by US Greentech) at 1.5 lbs per square foot over raw, round sand at 4.5 lbs per square foot.
 - 5) Coconut husks and fibers combined with rounded walnut shells (Natural Play by Shaw Sports Turf).
 - 6) Coconut fiber (Natural Cool by Greenplay).
 - 7) Olive pits (Elia Renufill by Safeplay, LLC).
 - 8) Mix of Olive Pits and Ecotherm (Thermoblend by Hellas).
 - 9) Or Approved Equal.

- c. Infill material not to exceed 75% of fiber height, Adjust infill ratios per fiber height of turf product being used while meeting required testing requirements.
 3. Pad System: Impact attenuation layer. The pad system shall be installed per manufacturer's specifications and instructions consisting of one of the following. Pad system shall be reusable through a minimum of two turf lifecycles. Turf shall be replaced with minimal damage to the pad system.
 - a. No natural, recycled, or EPDM rubber materials. No poured in place systems.
 - b. An interlocking tile or rolled pad system installed per manufacturer's specifications and instructions consisting of one of the following:
 - 1) Thermal bonded, closed cell cross-linked polyethylene foam (Schmitz ProPlay-20, minimum 20mm).
 - 2) Expanded polypropylene composite (Brock SP17 Shock Pad, Brock YSR Shock Pad).
 - 3) Expanded polypropylene (Safeplay Soteria Max).
 - 4) Polyolefin composite (En Plast ShockDrain KBA 780).
 - c. Thickness to provide the required performance requirements (15mm minimum).
 - d. Or Approved Equal
 4. The total system (fabric, infill & pad) will meet the minimum performance requirement and characteristics:
 - a. Head Injury Criterion (HIC): 800 HIC / 1.3-meter drop height at install and less than 1000 HIC at end of turf warranty period.
 - b. The G-Max Value throughout the warranted lifecycle of the synthetic turf playing surface is not to exceed 165 / 61 cm drop height.
 - c. Relative Abrasive Index Minimum: 10 +/- 2; Maximum 20 +/- 2
 - d. Average Ball Rebound Height Minimum: 20"; Maximum 30"
 - e. Coefficient of Restitution (CR) Minimum: 0.35 (CR); Maximum: 0.45 (CR)
 - f. Average Ball Bounce Minimum: 30"; Maximum: 40"
 - g. Soccer Shoe Traction Minimum - Dry: Static COF 1.30 – Dynamic COF 0.85; Maximum - Dry: Static COF 1.60 – Dynamic COF 1.15
 - h. Soccer Shoe Traction Minimum - Wet: Static COF 1.30 – Dynamic COF 0.85; Maximum - Wet: Static COF 1.60 – Dynamic COF 1.15
 - i. Football Shoe Traction Minimum - Dry: Static COF 1.30 – Dynamic COF 0.85; Maximum - Dry: Static COF 1.60 – Dynamic COF 1.15
 - j. Football Shoe Traction Minimum - Wet: Static COF 1.30 – Dynamic COF 0.85; Maximum - Wet: Static COF 1.60 – Dynamic COF 1.15
 5. Glue, thread, paint, seaming fabric and other materials used to install the synthetic turf system per the turf manufacturer. Ultrabond Turf PU 1K or equal.
 6. Line Markings, Numbers and Logos: All markings should be tufted in-place, inlaid or glued.
 - a. Inlay gaps should be uniform. The gap between the fibers should not exceed the gauge of the tufting.
 - b. Lines and markings must conform to the appropriate association/organization guidelines for the intended level of use.
 7. Base Layer: Minimum 3" layer of ¼" to ¾" crushed rock compacted to 90% proctor density. Base layer must drain. Use finer material as a top dressing between larger rock and pad.
 8. Drainage system (if required), including pipe, trenching fill and other materials.
- B. The installed synthetic turf shall meet all mandatory testing requirements. All testing to be conducted by independent third-party testing laboratory. Contractor must submit required testing. See Paragraph 1.4.
- C. The carpet shall consist of fibers tufted into a primary backing with a secondary backing.
 1. The fibers/carpet shall be furnished in 15' wide rolls. Rolls shall be long enough to go from sideline to sideline without splicing. Head seams, other than at sidelines, will not be acceptable
 2. The fiber tufts shall be fanned or unfolded prior to installation, rolling or spiraling is not acceptable.

- D. Thread for sewing seams of turf shall be as recommended by the synthetic turf Manufacturer. The method and adhesives used for inlaid markings shall be as recommended by the synthetic turf manufacturer.

2.3 FIELD GROOMER

- A. Supply a field groomer as recommended by the turf manufacturer to Owner. Include a towing mechanism compatible with a field utility vehicle and instructions.
- B. Acceptable Groomers:
 - 1. Geo-Groomer
 - 2. Greens Groomer 920SDE
 - 3. TurfKing SMG TK1205
 - 4. TurfCare TCA2000

PART 3 EXECUTION

3.1 GENERAL

- A. The installation shall be performed in full compliance with approved Shop Drawings.
- B. Only trained technicians, skilled in the installation of athletic caliber synthetic turf systems working under the direct supervision of the approved installer supervisors, shall undertake any cutting, sewing, gluing, shearing, topdressing or brushing operations.
- C. The designated Supervisory personnel on the project must be certified, in writing by the turf Manufacturer, as competent in the installation of this material, including sewing or gluing seams and proper installation of the Infill mixture.
- D. All designs, markings, layouts, and materials shall conform to all currently applicable National Collegiate Athletic Association rules and other standards that may apply to this type of synthetic grass installation.

3.2 EXAMINATION

- A. Verify that all sub-base, drainage and leveling is complete prior to installation.
- B. The surface to receive the synthetic turf shall be inspected by the Installer prior to beginning installation. The Installer shall make any adjustments to the sub-base needed to meet the compaction, planarity and permeability requirements of the synthetic turf manufacturer written guidelines. The surface must be perfectly clean as installation commences and shall be maintained in that condition throughout the process.
- C. The compaction of the aggregate base shall be 90%, according to the Modified Proctor procedure (ASTM D1557), and the surface tolerance-to-grade shall not exceed 0-1/4 inch over 10 feet and 0-1/2" from design grade.
- D. Correct conditions detrimental to timely and proper completion of Work.
- E. Do not proceed until unsatisfactory conditions are corrected.
- F. Beginning of installation means acceptance of existing conditions.

3.3 PREPARATION

- A. Prior to the beginning of installation, inspect the sub-base for tolerance to grade.

- B. Sub-base acceptance shall be subject to receipt of test results (by others) for compaction and planarity that sub-base is in compliance with manufacturer's specifications and recommendations.
- C. Dimensions of the field and locations for markings shall be measured by a registered surveyor to verify conformity to the specifications and applicable standards. A record of the finished field as-built measurements shall be made.
- D. When requested by Architect, installed sub-base shall be tested for porosity prior to the installation of the slit-film turf. A subbase that drains poorly is an unacceptable substrate.

3.4 INSTALLATION

- A. Install in accordance with Manufacturer's instructions. Infill materials shall be approved by the Manufacturer and installed in accordance with the Manufacturer's standard procedures.
- B. The full width rolls shall be laid out across the field. Turf shall be of sufficient length to permit full cross-field installation from sideline to sideline. No head or cross seams will be allowed in the main playing area between the sidelines. Utilizing standard state of the art sewing or gluing procedures each roll shall be attached to the next.
- C. Glued seams shall have glue continuous throughout the length and width of the seam overlap (per the manufacturer's standard procedures). Seams shall be flat, tight, and permanent with no separation or fraying.
- D. Sewn seams shall use a double thread/stitch method (per the manufacturer's standard procedures). Seams shall be flat, tight, and permanent with no separation or fraying.
- E. Infill materials shall be applied in numerous thin lifts. The turf shall be brushed as the mixture is applied. The infill material shall be installed to a depth determined by the Manufacturer.
- F. The Infill materials shall be installed to fill the voids between the fibers and allow the fibers to remain vertical and non-directional.
- G. Prior to the application of any line painting the turf shall be fibrillated by means of a nylon rotary brush to provide the look, feel, and safety of optimally maintained natural grass, including subtle undulations normally associated with natural grass athletic fields.
- H. Non-tufted or inlaid lines and markings shall be painted according to the recommendations of the turf Manufacturer and of the paint manufacturer. Several applications may be required.
- I. Synthetic turf shall be attached to the perimeter edge detail in accordance with the Manufacturer's standard procedures.
- J. Treat completed turf system with fabric softener per manufacturer's recommendations to remove static electricity.

3.5 CLEANING

- A. Protect installed synthetic turf from subsequent construction operations.
- B. Do not permit traffic over unprotected surface.
- C. Contractor shall provide the labor, supplies, and equipment as necessary for final cleaning of surfaces and installed items.

- D. All usable remnants of new material shall become the property of the Owner.
- E. The Contractor shall keep the area clean throughout the project and clear of debris.
- F. Surfaces, recesses, enclosures, etc., shall be cleaned as necessary to leave the work area in a clean, immaculate condition ready for immediate occupancy and use by the Owner.

END OF SECTION

SECTION 32 18 23.33

RUNNING TRACK SURFACING

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Construction and surfacing of outdoor running track. Work of this section includes.
 - 1. Track survey and layout
 - 2. Rough and finish grading
 - 3. Concrete curb and drainage system
 - 4. Application of synthetic track surface including event lines and marks.

1.2 RELATED SECTIONS

- A. Section 01 45 00 – Quality Control
- B. Section 31 22 13 – Grading
- C. Section 31 23 23.13 – Backfilling
- D. Section 03 30 00 – Cast-In-Place Concrete

1.3 REFERENCES

- A. Uniform Standard Specifications for Public Works' Construction, Offsite improvements, Clark County Area, Nevada (Latest Edition) herein referred to as Uniform Standard Specifications.
- B. American Sports Builders Association Track Construction Manual.

1.4 SUBMITTALS

- A. Submit under provisions of Section 01 31 00.
- B. Successful Track Oval Measurements Compliance Testing.
- C. Shop drawings: Provide layout drawings including all lane and event markings, dimensions, radiuses, event distances and details of all edge and joint treatment.
- D. Product Data: Provide manufacturer's data on complete surfacing system and its major components.
- E. Samples: Submit actual color samples of track material and line marking paint for color selection by Owner.
- F. Samples: Provide four (4) 12" x 12" samples of selected color material with lane marking paint applied.
- G. Manufacturer's Instructions: Provide manufacturer's installation instructions indicating required preparation work, sub-base procedures, installation procedures and environmental limitations. Provide manufacturer's maintenance instructions.
- H. Certificates: Provide manufacturer's certificates of compliance indicating conformance with specified requirements.

- I. Surfacing Mix Design: Provide the job mix formula, by weight and volume, of the rubber/ urethane mix and weight of the cured material per square yard.
- J. Paint source and technical data on the paint.

1.5 QUALITY ASSURANCE

- A. All Work of this section shall be supervised by a American Sports Builders Association member with appropriate experience with the specified track surface system.
- B. Contractor for the track surfacing shall be a member of the American Sports Builders Association. Athletic surfacing shall be installed under the direct, full-time supervision of manufacturer's authorized representative.
- C. The placement of the asphalt surface course shall be under the direct supervision of track builder, who is recognized be the American Sports Builders Association.
- D. Track Oval Measurements Compliance Testing- Provide the owner Certification that the track markings meet National Federation of State High School Associations (NFHS) and Nevada Interscholastic Athletic Association current standards. The certification shall be completed, stamped, and signed by an independent licensed surveyor or engineer and the surveyor or engineer shall have provided similar work for at least 5 running tracks in the last 5 years.

1.6 QUALIFICATIONS

- A. Manufacturer: Manufacturer of surfacing system to be an established firm experienced in the manufacture of outdoor sport surface system with a minimum of five (5) years experienced in the southwestern U.S. All materials of the system shall be the product of one manufacturer.
- B. Installer:
 - 1. Installer to be a firm experienced in the installation of outdoor sport surface systems over recommended base materials. Installer shall have a minimum of five (5) years documented experience in running track construction.
 - 2. Installer to be recognized and approved by the outdoor sport surface system manufacturer.
 - 3. The owner reserves the right to approve or disapprove the contractor's qualification and ability to perform the work of this Section.
 - 4. Membership in the American Sports Builders Association is required. It is preferable for the Contactor to have a Certified Track Builder on staff for this project and furnish proof at the pre-bid conference.

1.7 PRE-INSTALLATION CONFERENCE

- A. Convene two weeks prior to the scheduled finish grading work, i.e., placement and compaction of the Type II base construction and placement of any curbing. Provide two weeks' written notice of time and date to Owner and Architect.
- B. General Contractor to coordinate this conference and inform the following persons, in writing, of the requirements for their attendance.
 - 1. Architect/Representative and Civil Engineer
 - 2. Paving Subcontractor
 - 3. Surfacing system manufacturer's technical representative
 - 4. Installer's technical representative.
 - 5. General Contractor's Superintendent
 - 6. Owner's Construction Contract Representatives
 - 7. Project testing laboratory of record

- C. Contractor to provide proof of experience of manufacturer and installer at the conference. Contractor shall provide copies of accepted submittals including mix design, product data, shop drawings and manufacturer's installation instructions.
- D. Attendees shall review all pertinent details, specifications, the geotechnical report (if appropriate) and referenced standards. Discussion shall include scheduling of work of this section in coordination with other grading and paving work on the project, paving and surfacing installation procedures as proposed by the synthetic surface subcontractor, source and delivery of specified bituminous materials and surfacing materials, review of previously approved submittals for these materials and the guaranteed requirements of this section.
- E. The Architect or Owner's Construction Representative will take minutes of this meeting and any required subsequent meetings, noting all discussion items and resolution/conclusions. Draft copies of the minutes will be furnished to all attendees for their review and input prior to issuance of finalized meeting minutes.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Comply with provisions of Section 01 60 00.
- B. Deliver packaged materials and store in original containers with seals unbroken and labels intact until time of use, in accordance with manufacturer's instructions.
- C. Area where materials are to be stored to be maintained at a minimum of 55°F and under 50% relative humidity.
- D. Protect work during and after installation, until acceptance by the Owner, and in accordance with the outdoor sport surface system manufacturer's printed instructions.
- E. Handle, store, mix and apply proprietary materials in compliance with manufacturer's instructions. Provide materials obtained from one source.

1.9 WARRANTY

- A. The general contractor shall complete the attached form and submit per provisions of Section 01 70 00 of the project closeout documentation.
- B. Upon project completion and prior to the release of final payment, the contractor shall execute a five (5) year warranty for materials and workmanship.

PART 2 PRODUCTS

2.1 RUNNING TRACK SURFACING SYSTEM:

- A. Manufacturers – Urethane Base Mat
 - 1. Beynon 200
 - 2. Hellas Construction Sport Track 200 Track Surfacing System.
 - 3. APT- Rekortan BS
 - 4. Stockmeier Urethanes – Stobitan SC
 - 6. Substitutions: Under provisions of Section 01 25 00.
- B. Urethane Base Mat:
 - 1. Polyurethane bonded rubberized tack surface shall be equal to the basic, black one-half (1/2) inch minimum thickness, single layer, permeable EPDM, sports surface for all track and field events.
 - 2. Polyurethane materials for the project shall be "MDI" (Mercury) free.
 - 3. Polyurethane shall be a two-part, single component urethane.

4. Polyurethane to be delivered in unopened containers. EPDM granules shall be clean and dry and shall be from 1.5 mm to 3 mm in size.
 5. Mixing shall be done in a consistent and uniform manner and approved by NCSD.
- C. Structural Spray:
1. Contractor shall provide for approval, a job mix formula of the rubber/urethane structural spray mix.
 2. Minimum for the rubber shall be 1.8 pounds per square yard, per coat.
 3. Mix ratio shall be 40 percent rubber and 60 percent urethane.
 4. Rubber: EPDM granules shall be clean and dry and shall be from .5mm to 1.5 mm in size, color red.
 5. Polyurethane: Red-pigmented 2 part, single component urethane equal to Qualipur 5050 by Advanced Polymer Technology Corp.
- D. Polyurethane bonded rubber surface shall completely cover all asphaltic paving as indicated on the plan.
- E. Prime coat for urethane surface shall be in accordance with manufacturer's specifications.

PART 3 EXECUTION

3.1 PREPARATION

- A. Track and events layout and dimensions shown on the drawings are for track and event locations in general. Final layout and dimensions shall be in accordance with American Sports Builders Association Track Construction Manual.
- B. Grading of track areas shall be in accordance with Section 31 22 13 – Grading
- C. Contractor shall ensure the curb and drainage system installed around the interior perimeter of the track is installed horizontally and vertically correct before proceeding with subgrade preparation. Contractor to have the curb surveyed by third party to ensure it is correctly installed. Contractor will correct the curb as necessary.
- D. Subgrade preparation and base course shall be completed in accordance with Section 31 23 23.13 – Backfilling.

3.2 FINAL SURFACE INSPECTION

- A. Prior to application of a synthetic surface system, the running track and field events shall be flooded with water and allowed to drain; then, if remaining water covers a five (5) cent piece in any area, that area shall be patched and leveled in accordance with recommendations of the manufacturer of the synthetic surface system specified. Note: If the standing water does not cover a five (5) cent piece after 45 minutes, it is considered within tolerance. Reflooding and patching may be necessary until "bird baths" are minimized.
- B. Contractor to have track surveyed by third party prior to the placement of the synthetic surface system to ensure longitudinal and lateral surface slope requirements are met. Areas that do not meet these requirements will be corrected to the owner's satisfaction by the contractor at no cost to NCSD.

3.3 SURFACING

- A. Applicator for the track surfacing shall review, approve and accept the final asphalt surfacing.
- B. All work shall be performed within the temperature ranges required by the manufacturer.
- C. All areas to be surfaced shall be cleaned using brooms, blowers and/or pressure washers, to the satisfaction of NCSD or their consultant.

- D. The final compacted surface of the asphalt shall meet the top of the curb on the inside of the track and meet the top of the asphalt on the outside of the track. The concrete curb and the outer perimeter of track shall be taped. The urethane surface shall be tapered to the tape. The taper shall be one inch wide.
- E. Entire surface area shall be primed with a compatible polyurethane primer as recommended by the manufacture.
- F. Apply a prime coat followed by two coats of the structural coating to the entire track and field events areas.

3.4 STRIPING

- A. Track shall be striped for a 400 meter, 8-lane track with 42 inch lanes. Paint running lines for all events in metric. All line painting and event location will be in accordance with the National Federation of State High School Associations and Nevada Interscholastic Athletic Association requirements to include the following and as noted on the drawings:
 - 1. Hurdle markings (for 100 & 110 meters) shall be marked in both directions in contrasting colors.
 - 2. 1-turn stagger box alley mini waterfall starts 2 lanes per alley.
 - 3. 1-turn, 2-turn, 3-turn and 4-turn staggers.
 - 4. Waterfall start for 1500 meter and 3000 meter.
 - 5. The inside lane line of Lane 1 only shall be yellow.
 - 6. Lane numbers, shall be marked in two locations.
- B. The paint shall be two component metallic latex by Sherwin Williams.
- C. The track layout shall have a minimum of 24 inches of synthetic surface beyond the painted lane along the interior and a minimum of 6 inches of synthetic surface beyond the painted line along the exterior.
- D. The Contractor shall furnish the Owner with a scaled as-built plan depicting the layout and measurements. The Contractor shall also furnish a manual of the track marking calculations.
- E. The Contractor shall make provisions for the painting of the school name/logo on the running track surface in two locations.

3.5 CLEANING

- A. Upon completion of work herein, clean all finish outdoor sport surfaces so they are free of foreign matter.
- B. Remove all excess and waste material from the area of work.
- C. Protect the finish work until approval for release by the installer and acceptance by the Owner.

3.6 DEMONSTRATION

- A. Once the school is occupied, the track surfacing contractor shall schedule an in-service training session for selected members of the school staff pertaining to the operation and maintenance of the running track.
- B. Items to be covered in this session include, but are not limited to, cleaning the track, warranty repair work, what not to do to the track surface, treatment of graffiti, periodic inspection and maintenance and sources of damage and how to minimize them.

END OF SECTION

RUNNING TRACK CONSTRUCTION

GUARANTEE

The Contractor, the installer and the Synthetic Surface System Manufacturer hereby jointly and severally guarantee the color system to be free from manufacturing and installation defects, guarantee adhesion of the surface to the underlying asphalt, and guarantee adhesion of the system for a period of five (5) years from the date of substantial completion of the project. Exclude from this guarantee is normal wear which will take place in the area.

We the undersigned will, at our expense, make or cause to be made such permanent repairs to or replace of such synthetic surface system having such fault or defect in any of the material or workmanship applied by or through the undersigned as many as may be necessary to restore or replace such running track without defect.

Location of Project: _____

Date of Substantial Completion _____

Date Guarantee Expires _____

IN WITNESS THEREOF, this instrument has been duly executed by:

GENERAL CONTRACTOR

By: _____
Signature of Authorized Officer

TITLE: _____

FIRM: _____

DATE: _____

SEAL

SYNTHETIC SURFACE MANUFACTURER

By: _____
Signature of Authorized Officer

TITLE: _____

FIRM: _____

DATE: _____

SEAL

SYNTHETIC SURFACE SYSTEM INSTALLER

By: _____
Signature of Authorized Officer

TITLE: _____

FIRM: _____

DATE: _____

SEAL

**SECTION 32 3113
CHAIN LINK FENCES AND GATES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Posts, rails, and frames.
- B. Wire fabric.
- C. Manual gates with related hardware.
- D. Accessories.

1.02 RELATED REQUIREMENTS

- A. Section 03 3000 - Cast-in-Place Concrete: Concrete anchorage for posts.
- B. Section 08 7100 - Door Hardware: Gate locking device.

1.03 REFERENCE STANDARDS

- A. ASTM A153/A153M - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware; 2023.
- B. ASTM A392 - Standard Specification for Zinc-Coated Steel Chain-Link Fence Fabric; 2011a (Reapproved 2022).
- C. ASTM F567 - Standard Practice for Installation of Chain-Link Fence; 2023.
- D. CLFMI CLF-SFR0111 - Security Fencing Recommendations; 2014.
- E. FS RR-F-191/1D - Fencing, Wire and Post Metal (Chain-Link Fence Fabric); 1990.

1.04 SUBMITTALS

- A. See Section 01 3000 - Administrative Requirements, for submittal procedures.
- B. Product Data: Provide data on fabric, posts, accessories, fittings and hardware.
- C. Shop Drawings: Indicate plan layout, spacing of components, post foundation dimensions, hardware anchorage, and schedule of components. See CLFMI CLF-SFR0111 for planning and design recommendations.

1.05 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section, with not less than three years of documented experience.
- B. Installer: Company specializing in installation of the products specified in this section with minimum three (3) years experience.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Chain Link Fences and Gates:
 - 1. Anchor Fence Co.
 - 2. American Fence Corp.
 - 3. United States Steel Corp.
 - 4. Allied Tube & Conduit, SS-40 Chain Link Fence Pipe.
 - 5. American Tube Company Inc.

2.02 COMPONENTS

- A. Line Posts: 1.9 inch (48 mm) diameter.
- B. Corner and Terminal Posts: 2.38 inch (60 mm) diameter.
- C. Gate Posts: 4 inch (101.6 mm) diameter.
- D. Top and Brace Rail: 1.66 inch (42 mm) diameter, plain end, sleeve coupled.

- E. Gate Frame: 1.88 inch (47.8 mm) diameter for welded fabrication.
- F. Fabric: 2 inch (51 mm) diamond mesh interwoven wire, 6 gauge, 0.1920 inch (4.9 mm) thick, top selvage knuckle end closed, bottom selvage twisted tight.
- G. Tension Wire: 6 gauge, 0.1920 inch (4.9 mm) thick steel, single strand.
- H. Tie Wire: Aluminum alloy steel wire.

2.03 MATERIALS

- A. Line Posts: Type I round in accordance with FS RR-F-191/1D.
- B. Terminal, Corner, Rail, Brace, and Gate Posts: Type I round in accordance with FS RR-F-191/1D.
- C. Wire Fabric: ASTM A392 zinc coated steel chain link fabric, Class II.
- D. Concrete: Type specified in Section 03 3000 Type 5, 3000 psi.

2.04 MANUAL GATES AND RELATED HARDWARE

- A. Hardware for Single Swinging Gates: 180 degree hinges, 2 for gates up to 60 inches (1,525 mm) high, 3 for taller gates; fork latch with gravity drop and padlock hasp; keeper to hold gate in fully open position.
- B. Hardware for Double Swinging Gates: 180 degree hinges, 2 for gates up to 60 inches (1,525 mm) high, 3 for taller gates; drop bolt on inactive leaf engaging socket stop set in concrete, active leaf latched to inactive leaf preventing raising of drop bolt, padlock hasp; keepers to hold gate in fully open position.

2.05 ACCESSORIES

- A. Caps: Cast steel galvanized; sized to post diameter, set screw retainer.
- B. Fittings: Sleeves, bands, clips, rail ends, tension bars, fasteners and fittings; steel.
- C. Extension Arms: Cast steel galvanized, to accommodate 3 strands of barbed wire, single arm, vertical.

2.06 FINISHES

- A. Components and Fabric: Vinyl coated over coating of 1.8 ounces per square foot galvanizing (over coating of 550 g/sq m galvanizing).
 - 1. Hardware: Hot-dip galvanized to weight required by ASTM A153/A153M.
 - 2. Accessories: Same finish as framing.
- B. Color(s): To be selected by Architect from manufacturer's standard range.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install framework, fabric, accessories and gates in accordance with ASTM F567.
- B. Place fabric on outside of posts and rails.
- C. Set intermediate posts plumb, in concrete footings with top of footing 2 inches above finish grade. Slope top of concrete for water runoff.
- D. Line Post Footing Depth Below Finish Grade: ASTM F567.
- E. Corner, Gate and Terminal Post Footing Depth Below Finish Grade: ASTM F567.
- F. Brace each gate and corner post to adjacent line post with horizontal center brace rail _____. Install brace rail one bay from end and gate posts.
- G. Provide top rail through line post tops and splice with 6 inch (150 mm) long rail sleeves.
- H. Install center brace rail on corner gate leaves.
- I. Do not stretch fabric until concrete foundation has cured 28 days.

- J. Stretch fabric between terminal posts or at intervals of 100 feet (30 m) maximum, whichever is less.
- K. Position bottom of fabric 2 inches (50 mm) above finished grade.
- L. Fasten fabric to top rail, line posts, braces, and bottom tension wire with tie wire at maximum 15 inches (380 mm) on centers.
- M. Attach fabric to end, corner, and gate posts with tension bars and tension bar clips.
- N. Install bottom tension wire stretched taut between terminal posts.
- O. Do not attach the hinged side of gate to building wall; provide gate posts.
- P. Install hardware and gate with fabric to match fence.
- Q. Provide concrete center drop to footing depth and drop rod retainers at center of double gate openings.
- R. Ground fence in accordance with Division 33.
- S. Install gate locking device specified in Section 08 7100.

3.02 TOLERANCES

- A. Maximum Variation From Plumb: 1/4 inch (6 mm).
- B. Maximum Offset From True Position: 1 inch (25 mm).
- C. Do not infringe on adjacent property lines.

END OF SECTION

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SECTION 32 84 00
LANDSCAPE IRRIGATION

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Pipe and fittings, valves, and accessories.
- B. Connection to utilities.
- C. Control system.

1.2 RELATED SECTIONS

- A. Section 32 18 13 – Synthetic Turf

1.3 REFERENCES

- A. American Society for Testing and Materials (ASTM), D1785 Standard Specifications for Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
- B. American Society for Testing and Materials (ASTM). D 2241 Polyvinyl Chloride (PVC) Plastic Pipe (SDR-PR).
- C. American National Standards Institute / American Society for testing and Materials (ANSI / ASTM). D 2564 Solvent Cement for Polyvinyl Chloride (PVC) Plastic Pipe and Fittings.
- D. American National Standards Institute / American Society for testing and Materials (ANSI / ASTM). D 2672 Standard Specification for Joints for IPS PVC Pipe Using Solvent Cement.
- E. American Society for Testing and Materials (ASTM). D 2855 Standard Practice for Making Solvent-Cemented Joints with Poly Vinyl Chloride (PVC) Pipe and Fittings.
- F. American National Standards Institute / American Society for testing and Materials (ANSI / ASTM). F 656 Standard Specification for Primers for Use in Solvent Cement Joints of Poly (Vinyl Chloride) (PVC) Plastic Pipe and fittings.
- G. American Society for Testing and Materials (ASTM), F 1498 Standard Specifications for Taper Pipe Threads 60° for Thermoplastic Pipe and Fittings.
- H. American Water Works Association (AWWA). C-900 and C909 Ring Tight Pipe and Fittings.
- I. Federal Specifications O-F-506 Flux, Soldering; Paste and Liquid.
- J. Federal Specifications A-A-58092 Tape, Antiseize, Polytetrafluorethylene.
- K. National Electric Manufacturer's Association (NEMA).

1.4 QUALITY ASSURANCE

- A. Attend a pre-construction meeting with the Owner's Representative and General Contractor prior to any work in this section.
- B. Regulatory Requirements, Codes, and Standards:
 - 1. Provide certificate of compliance from authority having jurisdiction indicating approval of products in system.
 - 2. Comply with requirements of utility supplying water for prevention of backflow and back siphonage.
- C. Installer Qualifications: Installer having minimum 5 years documented experience in irrigation systems similar in material, design, and extent to work indicated and having a successful record of service performance. Licensed in the state of Nevada with a C-10 license.
- D. Listing/Approval Stamp, Label, or Other Marking: On equipment, specialties, and accessories made to specified standards.
- E. Listing and Labeling: Equipment, specialties, and accessories listed and labeled in National Electrical Code, Article 100.
 - 1. Listing and Labeling Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL) as defined in OSHA Regulation 1910.7.
- F. Product Options: Irrigation system piping, specialties, and accessories are based on specific types, manufacturers, and models indicated. Components with equal performance characteristics produced by other manufacturers may be considered, provided deviations in dimensions, operation and other characteristics do not change design concept or intended performance as judged by the Owner. The burden of proof of product equality is on the Contractor.
 - 1. All piping, fittings, components, and accessories to be of domestic manufacture.

1.5 SUBMITTALS

- A. Submit under provisions of Division 1.
- B. Contractor shall submit irrigation components to be used on project, including pipe and valves.
- C. Manufacturer's Installation Instructions.

1.6 PROJECT RECORD DOCUMENTS

- A. Submit under provisions of Division 1.
- B. Project Record Drawings:
 - 1. At the onset of the irrigation, the installation Contractor shall provide at least one full-size set of original irrigation plans. Make copies of the plans as necessary to maintain clean, legible As-Built plans for the Project. At the end of every day in which work is done on the irrigation installation, revise the as-built plans to include the Work accomplished that day in red ink. A copy of these Record Drawings, with red ink information, shall be available for review by Owner at Project Site at all times. Dimension from two permanent points of reference (building corners, sidewalk, road intersections or permanent structures), location of following items:
 - a. Connection to existing water lines.
 - b. Pressurized Mainline (dimension every 50 feet along routing).
 - c. All Sleeves (both ends).

- d. Irrigation valves: including control valves, gate valves and quick coupling valves.
- e. Control wire routing.
- f. All wire splices.
- g. Other related equipment as directed by Owner's Representative.
- 2. Owner can choose not to certify any pay request submitted by the Contractor if the Record Drawings are not current at the end of each week, and processing of pay request can be withheld until records are updated.
- 3. At the end of Project, before substantial completion is issued, all Record Drawings shall be submitted to Owner.

1.7 OPERATION AND MAINTENANCE DATA

- A. Submit under provisions of Division 1.
- B. Provide instructions for operation and maintenance of system and controls and manufacturer's parts catalog.
 - 1. Valves, including Isolation Valves and Quick Coupling Valves

1.8 REGULATORY REQUIREMENTS

- A. Conform to applicable code for piping and component requirements.

1.9 FIELD MEASUREMENTS AND VERIFICATIONS

- A. Performance of The Work at each site shall not proceed until the Owner and Contractor have walked the site and agree on the exact scope of work.
- B. Contractor shall verify all existing conditions before preparation of Shop Drawings. It is the Contractor's responsibility to verify and include requirements for any extensive work prior to preparation of Shop Drawings.
- C. Current water pressure is to be checked at the point of connection for the irrigation system. Verify that a sufficient PSI is available for proper function of quick coupling valves and of any hose attachments to be used. Bring inadequacies to the Owner prior to the start of work.

1.10 DELIVER, STORAGE, and HANDLING

- A. Deliver, unload, store, and handle materials, packaging, bundling, and products, in dry, weatherproof condition to prevent damage, breakage, deterioration, intrusion, ignition, and vandalism. Deliver in original unopened packaging containers prominently displaying manufacturer name, volume, quantity, contents, instructions, and conformance to local, state, and federal law. Remove and replace cracked, broken, or contaminated items or elements prematurely exposed to moisture, inclement weather, snow, ice, temperature extremes, fire, or jobsite damage.
- B. Handling of PVC Pipe: Exercise care in handling, loading and storing of PVC pipe. All PVC pipe shall be transported in a vehicle that allows length of pipe to lie flat so as not to subject it to undue bending of concentrated external loads. All sections of pipe that have been dropped, dented, or damaged shall be discarded, and if installed, shall be removed and replaced with new piping.

1.11 PROJECT CONDITIONS

- A. Perform site survey, obtain NCSD Record Drawings, research public utility records, and verify existing utility locations. Verify irrigation system piping may be installed in compliance with referenced standards.

- B. Contractor shall measure the pressure of the existing irrigation.
- C. Contractor shall verify the functionality of the existing irrigation system.
- D. Protection of Property:
 - 1. Preserve and protect all trees, monuments, structures, and paved areas from damage due to Work of this Section. In the event damage does occur, all damage to inanimate items shall be completely repaired or replaced to satisfaction of Owner. All injury to trees, shrubs, and ground covers shall be repaired by Contractor, and all costs of such repairs shall be charged to and paid by Contractor.
 - 2. Protect buildings, walks, walls, landscaping, irrigation system, and other property from damage. Flare and barricade open ditches. Damage caused to asphalt, concrete, landscaping, irrigation system, or other building materials surfaces shall be repaired or replaced at no cost to Owner. Restore disturbed areas to original condition.
- E. Existing Trees
 - 1. All trenching or other Work under limb spread of any and all evergreens or low branching deciduous material shall be done by hand or by other methods so as to prevent damage to limbs or branches.
 - 2. Where it is necessary to excavate adjacent to existing trees, use all possible care to avoid injury to trees and tree roots. Excavation, in areas where 2-inch and larger roots occur, shall be done by hand. Roots 2-inches or larger in diameter, except directly in the path of pipe of conduit, shall be tunneled under and shall be heavily wrapped with burlap to prevent scarring or excessive drying. Where a trenching machine is operated close to trees having roots smaller than 2-inches in diameter, a wall or trench adjacent to tree shall be hand trimmed, making clean cuts through roots. Roots 1-inch and larger in diameter shall be painted with two coats of "Tree Seal". Trenches adjacent to trees shall be closed within 24 hours, and when this is not possible, side of trench adjacent to tree shall be kept shaded with moistened burlap or canvas.
- F. Protection and Repair of Underground Lines:
 - 1. Request proper utility company to stake exact location (including depth) of all underground water, electric, gas, or telephone lines. Take whatever precautions are necessary to protect these underground lines from damage. In the event damage does occur, all damage shall be repaired by Contractor, and all costs of such repairs shall be paid by Contractor unless other arrangements have been made.
- G. Replacement of Paving and Curbs: Where trenches and lines cross existing roadways, paths, asphalt, concrete, curbing, and other surfaces - damage to these shall be kept to a minimum and shall be restored to original condition, using cuts parallel and perpendicular to existing surfaces and adjacencies. Repairs and installations shall be approved by Owner.
 - 1. Repair to damages to the existing track surfacing shall be coordinated with Owner. Direction for repair must be approved by Owner's Representative in writing.
- H. Cleaning: Maintain continuous cleaning operation throughout duration of Work. Dispose of all trash or debris generated by installation of irrigation system off-site at no additional cost to Owner.

1.12 SEQUENCING AND SCHEDULING

- A. Maintain uninterrupted water service to building during normal working hours. Arrange for temporary water shutoff with Owner as necessary.

- B. If irrigation to existing campus must be turned off, temporary irrigation or hand watering may be necessary. Coordinate with Owner's Representative. Temporary irrigation methods and watering schedules must be approved by Owner's Representative.
- C. Coordinate work with other trades.

1.13 EXTRA MATERIALS

- A. Provide the following extra components under provisions of Section 01 78 43.
 - 1. Two (2) matching quick coupler keys with swivels.
 - 2. Provide one (1) tool to open valve boxes.
 - 3. Provide one (1) key for each size of gate valve nut, if gate valves are installed. Provide proper length to access valve, gate valve handle is to extend three (3) feet above finish grade.

PART 2 PRODUCTS

2.1 MANUFACTURERS

- A. Products: valves, quick couplers, etc.. The Owner's Representative must approve any substitutions to the products.
- B. Hunter or Rain Bird: Quick coupling valve
- C. Nibco, Hammon, or Milwaukee: All brass gate valve, to match pipe size.
- D. Napac, Clow, or Nibco: All iron gate valve, to match pipe size.

2.2 MATERIALS

- A. General Piping:
 - 1. psi Supply Line to pump (from point of connection through backflow prevention unit). AWWA Standard C-900 or C-909, Class 150, 150 psi minimum, DR-18 pressure rating pipe, ASTM Standard D1784. Provide 2 feet 0 inches (minimum) of Class 51 Ductile Iron, 150 psi operating pressure rating, 250 psi surge pressure rating, before and after the piping has gone into the ground at the booster pump. Verify point of connection distance from booster pump enclosure and provide piping as needed between the two points.
 - 2. Pressure Supply Lines (downstream of backflow prevention units)
 - a. Schedule 40 PVC (1" – 3") rigid, unplasticized polyvinyl chloride (PVC) 1120, 1220 National Sanitation Foundation (NSF) approved pipe extruded from material meeting the requirements of Cell Classification D12454-A or 12454-B, ASTM Standard D1784 and D1785, with an integral belled end.
 - b. AWWA Standard C-900 or C-909, Class 150, 150 psi minimum, DR-18 pressure rating pipe, ASTM Standard D1784, with an integral belled end, for 4 inch and larger sizes. The joint design shall meet ASTM D3139 Joint qualification performance testing requirements, with the elastomeric seals (Gaskets) manufactured in compliance with ASTM F477.
 - 3. Lateral Piping:
 - a. Schedule 40 PVC rigid, unplasticized polyvinyl chloride (PVC) 1120, 1220 National Sanitation Foundation (NSF) approved pipe extruded from material meeting the requirements of Cell Classification D12454-A or 12454-B, ASTM Standard D1784 and D1785, with an integral belled end.
 - 4. Sleeving:

- a. Schedule 40 PVC (6" and smaller) rigid, unplasticized polyvinyl chloride (PVC) 1120, 1220 National Sanitation Foundation (NSF) approved pipe extruded from material meeting the requirements of Cell Classification D12454-A or 12454-B, ASTM Standard D1784 and D1785, with an integral belled end.
 - b. SDR 26, Class 200 PVC (8" and larger) rated at 160 PSI, rigid, unplasticized polyvinyl chloride (PVC) 1120, 1220 National Sanitation Foundation (NSF) approved pipe extruded from material meeting the requirements of Cell Classification D12454-A or 12454-B, ASTM Standard D1784 and D1785, with an integral belled end.
 - c. Install separate sleeve beneath paved areas to route each run of irrigation pipe or wiring bundle.
 - d. Sleeving diameter: two sizes larger than the piping inside, equal to twice that of the wiring bundle.
- B. Copper Pipe and Fittings:
1. Copper Pipe - Type K, hard tempered, in accordance with ASTM B4284.
 2. Joints - Soldered with solder, 45% silver, 15% copper, 16% zinc, and 24% cadmium and solidus at 1125oF and liquids at 1145oF.
 3. Copper tube fittings: ASME B16.22, wrought copper or cast-brass, solder joint, pressure type.
 4. Copper Unions: ASME B16.18, cast-copper-alloy body, hexagonal stock, with ball- and-socket joint, metal-to-metal seating surfaces, and solder-joint, threaded or solder-joint, and threaded ends.
 - a. Threaded Ends: Threads conforming to ASME B1.20.1.
5. Cast-Bronze Flanges: ASME B16.24, Class 150, raised ground face, bolt holes spot faced.
- C. Bronze Pipe and Fittings:
1. All brass pipe shall be rated Schedule 40 or greater in accordance with ASTM B-43. Brass pipe nipples will be in accordance with ASTM B-687.
 2. Treaded fittings for bronze pipe will be Schedule 40 in accordance with ASTM B-62-90.
- D. Plastic Pipe and Fittings:
1. Identification Markings:
 - a. Identify all pipe with following indelible markings:
 - 1) Manufacturer's name.
 - 2) Nominal pipe size.
 - 3) Schedule of class.
 - 4) Pressure rating.
 - 5) NSF (National Sanitation Foundation) seal of approval.
 - 6) Date of extrusion.
 2. Solvent Weld Pipe - Manufactured from virgin poly vinyl chloride (PVC) compound in accordance with ASTM D2241 and STM D1784; cell classification 12454-B, Type 1, Grade 1.
 - a. Fittings - Standard weight, Schedule 40, injection molded PVC; complying with ASTM D1784 and D2466, cell classification 12454-B.
 - b. Threads - injection molded type (where required).
 - c. Tees and Ells - Side gated.
 - d. Fittings - Schedule 80, injection molded PVC; complying with ASTM D1785 and D2467, cell classification 12454-B.
 - e. Threaded Nipples - ASTM D2464, Schedule 80 with molded threads.
 - f. Joint Cement and Primer - (ANSI/ASTM D2564).
 3. Ring tight PVC Pipe Fittings: Epoxy Coated Ductile Iron Grade 65-45-12, gasketed, deep bell with joint restraints for change of direction in mainline on 4 inch and larger pressure pipe only. Do not use for connection of valves to mainline. Reducers and plugs to be provided with bolt-on links. All fittings to have four lugs to accommodate joint restraints in any

- orientation. EBAA, Napac, Tyler, Sigma, Waterous or equal. Provide joint restraints on fittings 4 inches and larger as per manufactures recommendations.
4. Dielectric Fittings: Assembly or fitting with insulating material isolating joined dissimilar metals to prevent galvanic action and stop corrosion. These devices are a combination of copper alloy and ferrous metal threaded-end solder-end types, matching piping system materials.
 - a. Dielectric Unions: Factory-fabricated, union assembly, designed for 250-psig (1725 kPa) minimum working pressure at 180 degrees F (82 degrees C). Include insulating material isolating dissimilar metals and ends with inside threads according to ASME B1.20.1.
 - b. Dielectric Flanges: Factory-fabricated, companion-flange assembly for 150 psig (1035 kPa) or 300 psig (2070 kPa) minimum pressure to suit system pressures.
 - c. Transition Fittings: Manufactured assembly or fitting, with pressure rating at least equal to that of system and with ends compatible to piping where fitting is to be installed.
 5. Joining Materials.
 - a. Refer to Basic Mechanical Materials and Methods for joining materials not included in this section.
 - b. Primer, IPS P-70 (ASTM 656) Provide for all solvent cemented connections
 - c. Solvent Cement, IPS 705 or 721 (ASTM 2564) clear medium bond for rigid PVC pipe under 3 inches.
 - d. Solvent Cement, IPS 711 (ASTM 2564) gray medium bond for main line or sleeving PVC pipe 3 inches or more.
 - e. Solvent Cement, IPS 795 (ASTM 2564) for all flexible PVC pipe such as the GPH Irrigation pipe.
 - f. Solder: ASTM B 32, Alloys Sn95 and E.
 - g. Gaskets and fasteners for metal and metal-to-plastic flanged joints: ASME B16.21, nonmetallic, asbestos-free, flat, 1/8-inch (3-mm) thickness gaskets and ASME B18.2.1, carbon steel bolts, butts and washers.
 - h. Gaskets for plastic flanged joints: Materials recommended by plastic pipe and fittings manufacturer.
 - i. PTFE sealing tape (Teflon® Tape) shall have a minimum thickness of 3.5 mil, minimum of 1.2 density, and meet with FED. MIL SPEC A-A58092.
 6. Gasketed End Pipe - Manufactured from virgin poly vinyl chloride compound In accordance with ASTM D2241 and ASTM D1874; cell classification 1254-B, Type 1, Grade 1.
 - a. Fittings – Ductile iron, AWWA C153, with push-on joints or mechanical joint. Connections from main line to control valves or any 2-inch and smaller pipe outlet, a bronze saddle with double stainless steel straps shall be used.
 - b. Gaskets - Factory installed in pipe and fittings, having a metal or plastic support within gasket or a plastic retainer ring for gasket.
 - c. Lubricant - As recommended by manufacturer of pipe fittings.
 7. Saddles on C900 or C909 irrigation main lines to have a wide stainless steel band, brass saddle with double bolts on each side. Ford Style 202BS Double Band Saddles for C900 PVC pipe, A.Y. McDonald Model 3846 Service Saddles with Wide Stainless Steel Band for C900 PVC pipe, or approved equal.
 8. Detectable Marking Tape 3-inches in width shall be installed 12 above all irrigation main lines.
 - a. Tape shall consist of a minimum 5.0 mil overall thickness; 100% virgin polyethylene; acid, alkaline and corrosion resistant.
 - b. Elongation properties shall be in accordance with ASTM D882-80A and shall be: 80% MD and 135%TD.
 - c. Tape shall have a minimum 20-gauge solid aluminum foil core, adhered to a 2.55 mil polyethylene backing.
 - d. Tape color shall be blue and the legend shall read “Caution Irrigation Line Buried Below”.

- e. Tape tensile strength shall be in accordance with ASTM D882-80A and be not less than 7,800 PSI.
- f. Tape shall remain flexible as measured by ASTM D671-76 pliable hand value.

2.3 VALVES

- A. Gate Valves:
 - 1. Gate Valve for 3/4 inch through 1-1/2 inch Pipe - Brass construction; solid wedge, IPS threads, and non-rising stem with brass butterfly operating handle. Nibco or equal.
 - 2. Nonrising Stem Gate Valves 2 inches and larger: AWWA C509, resilient seated; bronze stem, cast iron, or ductile iron body and bonnet, stem nut, 200 psi working pressure; and ends that fit NPS dimensions, PVC pipe. Include elastomeric gaskets. Napac, Nibco, Clow, or equal. Gate Valves 2 inches or larger shall have a minimum of a ten (10) year warranty.
 - 3. Valves under 2 inch shall be threaded, valves 2 1/2 inch to 3 inch shall be flanged with stainless steel bolts and nuts, valves 4 inch and larger shall be flanged with stainless steel bolts and nuts for above grade applications and push-on right tight with joint restraints for below grade applications.
- B. Valve Box and Cover for valves other than the rotor assembly: Carson, Dura Plastics, or Ametek rectangular high-density box, with locking equipment provided, use extension when depths are greater than the box and the valve being covered. Use green colored valve boxes in grass areas and sand colored valve boxes in shrub areas.
- C. Quick Coupling Valves: Factory fabricated, two-piece assembly, with non-potable purple cap. Include coupler water seal valve with ASME B1.20.7, 3/4-11.5NH threads for garden hose on outlet; and operating key.

PART 3 EXECUTION

3.1 INSTALLERS

- A. Nevada licensed irrigation contractor shall have C-10 license. See Section 1.4, C.
- B. The contractor shall have a Certified Irrigation individual on the site during the installation. See Section 1.4, C.
- C. The Contractor shall have readily available a copy of related specifications and plans on site at all times.

3.2 EXAMINATION

- A. Verify that field conditions are acceptable and are ready to receive work.
- B. Verify location of existing utilities and existing irrigation components.
- C. Verify that required utilities and services are available, in proper location and ready for use. If discrepancies exist bring them to the Owner's Representative immediately.
- D. Obtain all necessary permits.
- E. Prior to irrigation installation all landscape grades are to be established and approved by the Owner's Representative.

- F. Beginning of installation means installer accepts existing conditions.
- G. Verify existing irrigation water pressure and submit verification to the Owner.

3.3 PREPARATION

- A. Irrigation Shop Drawings. All materials, equipment, etc. are to be located in landscape areas. Any discrepancies between site conditions and approved Shop Drawings are to be brought to the attention of the Owner.
- B. Route piping to avoid plants and structures.
- C. Prior to the installation of sleeves, review layout requirements with other affected Work. Coordinate locations of sleeves under paving to accommodate system. Separate sleeves must be installed for the mainline, each lateral and all control wiring. Sleeves should be Schedule 40 (6 inch and smaller) or Class 200 PVC (larger than 6 inch), and sized minimum two sizes larger than the size of the irrigation pipe. Wire sleeves shall be a minimum of 2-inch Schedule 40 PVC. Compact backfill around sleeves to 95% Standard proctor Density within 2% of optimum moisture content in accordance with ASTM D1557. Extend sleeves 12-inches into all planting areas – past hardscape surfaces.
- D. Set stakes to identify proposed rotor assembly locations for approval by Owner prior to installation. Obtain Owner's Representative approval before excavation.

3.4 TRENCHING

- A. Trenching: Trench excavation shall follow, as much as possible, the Shop Drawings and shall be located outside the limit of play if possible.. Dig trenches straight and support pipe continuously on bottom of trench. Trench bottom shall be clean and smooth with all rock and organic debris removed. Pressure supply line trenches shall be over-excavated as required to allow for bedding material. Trench depth shall be uniform as required to meet minimum depth requirement for type of piping.
 - 1. Clearances:
 - a. Piping 3-Inches and Larger- Make trenches of sufficient width (14-inches minimum) to properly assemble and position pipe in trench. Minimum clearance of piping 3-inches or larger shall be 5-inches horizontally on both sides of the trench.
 - b. Piping Smaller than 3-Inches – Trenches shall have a minimum width of 10-inches.
 - c. Line Clearance – Provide not less than 6-inches of clearance between each line, and not less than 12-inches of clearance between lines of other trades.
 - d. Trench to accommodate grade changes.
 - 2. Pipe and Wire Depth:
 - a. Pressure Supply Piping – 24-inches to 30-inches deep to top of pipe.
 - b. Non-pressure Piping (rotor and pop-up) – 15-inches to 18-inches deep to top of the pipe.
 - c. Control wiring – 24-inches to 30-inches deep to top of the wire.
 - d. Communications Cable – 24-inches to 30-inches deep to top of the cable.
- B. Trench to accommodate grade changes.
- C. All piping is to be installed in an individual trench; no other piping is to be placed in this trench. Wiring can be in the same trench and should run on the side.
- D. Mainline piping is to be installed in an individual trench; no other piping is to be placed in this trench. Wiring can be in the mainline trench and should run on the side of the pipe only.

- E. Do not stack piping in same trench, piping to have a 12 inches minimum separation when side-by-side. Where pipes cross above or below another pipe, provide a minimum separation of 6 inches.
- F. Backfilling: Do not begin backfilling operations until required system tests have been completed. Backfill shall not be done in freezing weather except with prior approval by Owner's Representative. Trenches shall be finished graded prior to walk-through of system by Owner's Representative.
 - 1. All pressure supply lines shall be bedded with construction grade sand 4-inches below invert of pipe, to 6-inches above top of pipe and width of trench.
 - a. Detectable Marking Tape 3-inches in width shall be installed 12 above all irrigation main lines.
 - 2. Materials – Excavated material is generally considered satisfactory for backfill purposes after completing bedding requirements. Backfill material shall be free of rubbish, vegetative matter, frozen materials, and stones larger than 1½-inches in maximum dimensions. Do not mix subsoil with topsoil. Material not suitable for backfill shall be hauled away. Contractor shall be responsible for providing suitable backfill if excavated material is unacceptable or not sufficient to meet backfill, compaction, and final grade requirements.
 - 3. Open excavations shall be protected in accordance with OSHA regulations.
 - 4. Compact backfill to 90% maximum density in 6-inch lifts, determined in accordance with ASTM D1557 utilizing the following methods:
 - a. Mechanical tamping.
- G. Piping Under Paving or Hard Surface areas:
 - 1. Provide for a minimum cover of 30-inches for all pressure piping and 24-inches for all non-pressure piping between the top of the pipe and the bottom of the aggregate base for piping installed under asphaltic concrete or concrete paving.
 - 2. Piping shall be bedded with construction grade sand or squeegee – 6-inches below pipe to 6-inches above pipe and width of excavation.
 - 3. Compact backfill material in 6-inch lifts at 95% maximum density determined in accordance with ASTM D1557 using manual or mechanical tamping devices.
 - 4. Set in place, cap, and pressure test all piping under paving, in presence of Owner's representative Approved Representative or Owner's representative prior to backfilling and paving operations.
 - 5. Piping under existing walks or concrete pavement shall be done by jacking, boring, or hydraulic driving, but where cutting or breaking of walks and/or concrete is necessary, it shall be done and replaced at no cost to Owner's representative. Obtain permission and prior approval to cut or break walks and/or concrete from Owner's representative.
- H. Do not backfill trenches until inspected by Owner's Representative.

3.5 INSTALLATION

- A. Install pipe, valves, controls, and outlets in accordance with manufacturer's instructions and all applicable codes.
- B. Water Supply and Point of Connection: Water supply shall be extended as shown from water supply lines.
- C. Provide separation between water lines and other utilities as required per local and applicable codes.
- D. Provide for thermal movement of components in system.
- E. After piping is installed open valves and flush system with full head of water.

- F. PVC Piping:
1. Snake pipe in trench as much as possible to allow for expansion and contraction, do not place mainline and lateral line in same trench. Place each line in a separate trench.
 2. When pipe laying is not in progress, or at end of each day, close pipe ends with tight plug or cap.
 3. Coordinate pressure supply line installation with required bedding operations.
 4. Stake all above grade PVC piping per details.
 5. Use 45° ells when making perpendicular crossing of above grade PVC piping, to depress bottom pipe.
 6. Lay pipe and make all plastic-to-plastic joints in accordance with manufacturer's recommendations.
 7. Install detectable marking tape 12-inches above all irrigation main lines.
- G. Solvent Welding:
1. PVC piping 4 inches and smaller, use IPS 705 or 721 cement
 2. Main line or sleeving PVC piping 3 inches and larger, use IPS 711 cement
 3. IPS flexible PVC piping (GPH Irrigation), use IPS-795 cement
 4. All PVC piping, including IPS flexible PVC pipe, use IPS P-70 primer
 5. All solvent welding shall be done in accordance to ASTM-2855
 6. NCSD retains the right to have the contractor remove up to five (5) different fittings for testing of proper solvent welding techniques at the contractors cost.
- H. Threaded Fittings:
1. All threaded fittings will use an approved threaded joint sealant.
 - a. PTFE sealing tape shall meet FED.MIL SPEC A-A58092 and installed per manufacturers' specifications and American Society for Testing and Materials (ASTM), F 1498.
 - b. Paste-type tread sealants must be compatible with the particular plastic used. Install per manufacturers' specifications and American Society for Testing and Materials (ASTM), F 1498. Spears 75TM thread sealant or NCSD approved equal.
- I. Gasketed End Pipes:
1. Lay pipe and make pipe to fitting or pipe to pipe joint, following 0R70 recommendations (Johns-Manville Guide for Installation of Ring-Tite Pipe), or pipe manufacturer's recommendations.
 2. All ductile iron fittings, flanges (PVC or Ductile Iron), joint restrains, valves, etc. to be installed underground shall be wrapped with a minimum 8 mil polyethylene clear sheeting and then secured with a 2 inch black, 10 mil polyvinyl chloride (PVC) tape with a high-tack adhesive provides a corrosion- and impact-resistant seal for all-weather protection, UV resistance and has a temperature rating of 176°F (80°C).
- J. Joint Restraints:
1. Joint restraints shall be installed on rubber gasketed pipe 4-inch diameter and larger at all bends 11 degrees and over, reducers, valves, and caps in accordance with pipe manufacturer's recommendations. Contact Owner's Representative and/or NCSD Civil and Landscape Quality Assurance Construction Inspector/Manager to visually inspect all joint restraints. Pressure test pressure supply line with joint restraints installed and exposed to view.
 2. Joint Restrain Harness:
 - a. Use a joint restraint harness, per manufacture's specifications, wherever joints are not positively restrained by flanged fittings, and/or threaded fitting.
 - b. Use a joint restraint harness with transition fittings between metal and PVC pipe or where extra support is required to retain a fitting or joint.

- c. Use bolts, nuts, retaining clamps, all-thread, or other joint restraint harness materials, which are zinc plated or galvanized.
3. Use joint restraints based on the following table:

Table of Joint Restraint Lengths for Slip Joint Pipe (150 psi working + 50% surge = 225 psi)							
Size	Minimum Restrained Length (Feet)						
	11°	22°	45°	90°	1D	2D	DE
4"	2	3	7	16	11	16	36
6"	2	5	10	23	24	32	50
8"	3	6	12	30	26	44	66
10"	4	7	15	36	26	45	80
12"	4	8	17	42	36	45	94

K. Valve Boxes

1. Install one valve box for each type of valve installed. Install Terra Bond Fabric around the outside of the valve box by taping. Extend fabric 12 inches beyond base of valve box, then backfill around the valve box and compact. Install gravel sump after compaction of all trenches. Place final portion of gravel inside valve box after valve box is backfilled and compacted. Valve boxes shall have locking equipment provided, be green in color when installed in turf areas, and tan in areas other than turf, use extension when depths are greater than the box and the valve being covered.
2. Top of valve boxes in synthetic turf areas shall have synthetic turf on valve box top that matches turf type and color and shall be flush with the adjacent synthetic turf.
3. Brand all valve box lids. Letter and number size shall be no smaller than 1-inch and no greater in size than 1 ½-inches. Depth of branding shall be no more than 3/8-inches into valve box lid as follows:
 - a. Wire splices – Brand all wire splice box lids with letters "WS".
 - b. Isolation Gate Valves – Brand all isolation gate valve box lids with letters "GV".
4. Standard 12 inches deep rectangular valve boxes with extension, when required, are to be used for electric solenoid valves and wire splices.
5. 10-inch round valve box is required for all gate valves, isolation valves, and drip flushing valves.
6. Valve boxes installed near any hard surface shall be parallel to the hard surface.
7. Valve boxes shall have Epic plastics valve box stabilizers or bricks under each corner of a rectangular valve box and minimum of three (3) under round valve boxes.

L. Control Wiring (as needed for repairs or as directed by Owner):

1. Low Voltage Wiring:
 - a. Bury control wiring between controller and electric valves in pressure supply line trenches, with wires consistently located below and to one side of pipe, on top of initial pipe bedding. Separate wire trenches will not be allowed unless approved by Owner's Representative prior to installation.
 - b. Bundle all 24 volt wires at 10-foot intervals with electrical or duct tape.
 - c. Provide an expansion loop at pressure supply line angle fittings, every electric control valve location (in valve box), and at minimum 500 feet intervals. Form expansion loop by wrapping wire at least 8 times around a ¾-inch pipe and with drawing pipe.
 - d. Make splices and electric control valve Connections using 3M direct bury splice DBY or DBR; or similar dry splice method provide 8-inch expansions loops.
 - e. Install control wire splices not occurring at control valve in a separate splice valve box.
 - f. Run all wire for future control valves to point coordinated with Owner. Coil a minimum of ten (10) feet at termination and install in 10: round valve box. Label all wires at termination.

- M. Ductile iron or stainless steel piping to be provided 2 feet - 0 inches minimum beyond the discharge side of booster pump. Ductile iron or stainless steel piping to be provided 2 feet - 0 inches minimum beyond the inlet side of booster pump including "Y" strainer and gate valve. Connection from water utilities to point of connection shall be C900 PVC piping (See Section 328400, Part 2, Pipe Materials). Extend from the backflow device with ductile iron a minimum distance of 2 feet - 0 inches horizontally once it has entered the ground.

3.6 FIELD QUALITY CONTROL

- A. Flush the new irrigation system in the football field. Flush the mainline and lateral lines after piping, risers, quick couplers, and valves are in place and connected, but prior to installation of sprinkler heads, thoroughly flush piping system under full head of water pressure until totally clean (15 minutes). Additional flushing will be required if piping is damaged or changes have occurred to the mainline system.
- B. Testing - Conduct tests in presence of Owner's Representative. Arrange for presence of Owner's Representative 48 hours in advance of testing. Pressure test is to be conducted prior to backfilling trench. Supply force pump and all other test equipment.
 - 1. After backfilling at key locations for stabilization of the mainline; all valves, quick couplers and other devices subject to constant line pressure; and fill pressure supply line with water. The main line is to be under designated static pressure for seven (7) continuous days before pressure test. Expose all couplings, fittings, and valves.
 - 2. Pressurize the new mainline in the football field to 40 PSI over the designated static pressure or 120 PSI, for a period of 2 hours.
 - 3. Leakage, Pressure Loss - Test is acceptable if no leakage or loss of pressure is evident during test period.
 - 4. Leaks - Detect and repair leaks.
 - 5. Retest system until test pressure can be maintained for duration of tests.
 - 6. Before final acceptance, pressure supply line shall remain under pressure for a period of 48 hours.
 - 7. Subsections of pressure piping may be tested independently to allow backfilling of trench, subject to review of Construction Inspector/Manager. However, a final pressure test must be conducted when the entire system is in place.
 - 8. After the main line pressure test is passed, if any comprise in the irrigation main line, the contractor will then have to re-test any section(s) that was comprised.
 - 9. NCSD reserves the right to require re-testing of the irrigation main line before the end of the warranty/maintenance period.

3.7 ADJUSTING

- A. ADJUSTING - Upon completion of installation, "fine-tune" entire system by performing the following:
 - 1. System is to be flushed one week prior to NCSD acceptance.
- B. Areas that do not conform to designated operation requirements due to unauthorized changes or poor installation practices shall be immediately corrected at no additional cost to the Owner. Verify all grass areas dimension prior to installation.

3.8 DEMONSTRATION

- A. Demonstrate to the Owner's Representative that the system meets coverage requirements, the controls function properly, operation of equipment, and accessories Review operating and maintenance information. Provide 7 days written notice in advance of demonstration.

3.9 WARRANTY

- A. Provide prime quality materials matching the originally installed equipment.
- B. Warrant irrigation materials, equipment, and workmanship against defects for a period of one (1) year from date of substantial completion.
 - 1. Fill and repair depressions.
 - 2. Restore landscape, hardscape or structural features damaged by settlement of irrigation trenches or excavations.
 - 3. Repair damage to premises caused by defective components.
 - 4. Make repairs within 24 hours of notification from Owner's Representative. If no response is made to the requested repair, the contractor(s) will be back-charged for costs that occur.
 - 5. Any work performed on the irrigation system during the warranty period is to be documented and extended with a full year warranty.
- C. The Contract Documents govern replacement materials, labor, and workmanship identically as with new work. Make replacements at no additional cost to Owner.
 - 1. The warranty applies to originally installed materials and equipment and replacements made during the warranty period.
 - 2. Expensed due to vandalism, before substantial completion and during maintenance period, shall be borne by Contractor.
 - 3. "Act of God" may become an acceptable reason for a warranty to be void. However, the contractor must make every reasonable attempt to prevent potential damage. "Act of God" damage will include unseasonable freeze, winds above 60 M.P.H., and flooding from excessive rain.
- D. Contractor shall accommodate operation on irrigation system as to not conflict with school hours and use of turf areas or other areas accessed by school.

END OF SECTION

SECTION 32 91 13

TRACK & FIELD EVENT INORGANIC MATERIAL MIX

PART 1 - GENERAL

1.1 SCOPE

- A. Furnish labor, material, and equipment necessary for the excavation, grading, and construction of the shot put and discus, and long jump and triple jump sand catchers as shown on the drawings and specified herein.

RELATED SECTIONS:

- A. Section 31 22 13 – Rough Grading
- B. Section 32 13 23.33 – Running Track Surfacing
- C. Section 32 23 16 - Excavation
- D. Section 32 84 00 – Landscape Irrigation

1.2 SUBMITTALS

- A. Submit under provisions of Section 01 33 00.
- B. Shot put and discus mix: The Contractor shall submit Shot put and Discus mix product information, including binder manufacturer and binder amount to be used with runway mix product(s). Submit 1 gallon sample of mix without binder. Mock-up with binder product data is required for review and approval. Mock-up is required for approval.
- C. A/E and Owner's Representative reserves the right to approve or reject materials, suppliers and subcontractors.
- D. Submit a labeled 5-pound sample of Long / Triple Jump Sand.
- E. Submit instructions to the A/E for continuing maintenance as part of the Operations and Maintenance Manual.
- F. Submit warranties, Paragraph 1.4

1.3 QUALITY ASSURANCE

- A. Provide 10'x10' mock-up at approved location of the discus and shot put runway mix installation for review and approval. Mockup shall have the discus / shot put edging on at least one side of the mock up.

1.4 WARRANTY

- A. Warranty all non-organic material and equipment for a period of one year from date of substantial completion.

1.5 EXTRA MATERIAL

A. Surface material

1. 50 lb. of the shot put / runway mix blended with proper amount of stabilizing binder additive, in protective bags.

PART 2 - MATERIALS

2.1 SHOT PUT AND DISCUS MIX

A. Discus and shot put runway shall be a mix of the following:

1. 70% - 80% 2mm to 3mm screened fines. Color: gold, tan or red.
2. 20% -30% cinder sand. Color: gold, tan or red.
3. Organic binder. Natracil, Stabilizer or approved equal.

B. Gradation for shot put & discuss:

PERCENT PASSING U.S STANDARD SIEVE

Sieve Size	#10	#40	#200
Percent Finer by weight	70-80	50-70	<30

C. Manufacturers

1. Mix per Contractor, approved through submittals and mock-up.
2. Pro Gold Shot Put Mix available through Gail Materials (951-667-6106).
3. Or approved equal.

2.2 LONG /TRIPLE JUMP SAND

- #### **A. Appearance:** Sand shall be washed, white pigmented, fluffy, non-crusting sand containing no organic material.
- #### **B. Particle Characteristics:** Sand must not be abrasive or stain.

C. Particle Size shall meet the following sieve:

Particle	Diameter	Sieve	% Retained by Weight
Gravel	>2mm	No. 10	≤ 2%
Very Coarse Sand	1.0 – 2.0 mm	No. 18	≤ 15%
Coarse Sand	0.5 – 1.0 mm	No. 35	78 – 100%
Medium Sand	0.25 – 0.5 mm	No. 18	78 – 100%
Fine Sand	0.1 – 0.25 mm	No. 140	78 – 100%
Very Fine Sand	0.05 – 0.10 mm	No. 270	≤ 5% Max.
Silt + Clay	< 0.05 mm	Pan	≤ 3% Max.

D. No particles larger than 2mm.

PART 3 - EXECUTION

3.1 SHOT PUT AND DISCUS MIX INSTALLATION

- A. Discus and shot put runway shall be a mix shall per plans and specifications.
- B. Organic binder. Use 8 to 10 pounds per ton of screened fines and cinder sand combination, or amount recommended by binder manufacturer for shot put and discus use. Material shall have a minimum swell volume of 35 ml/gm.
- C. Materials shall be mixed together into a homogenous mixture very rapidly with a pug mill or similar mixing machine, as recommended by binder manufacturer and installed in lifts. Apply water to wet entire lift of material, and compact each lift.
- D. Depth of mix: minimum 4" depth, maximum 5" depth, compacted.
- E. Install pre-emergent herbicide to areas prior to installation of the runway mix.

3.2 CLEANUP

- A. Keep areas of work clean, neat and free of litter or deleterious materials. Remove waste materials, tools and equipment and leave site in an orderly manner for other site operations.

3.3 MAINTENANCE

- A. Keep the shot put and discus level and weed free from invasive weeds. Keep this zone free from the Bermuda rhizome encroachment.
 - 1. Apply herbicide along the perimeter edge when rhizome encroachment takes place.
 - 2. Apply two additional applications of pre-emergent across the interior infields during the maintenance period.

END OF SECTION

SECTION 32 96 50

INVASIVE PLANT REMOVAL

PART 1 - GENERAL

1.1 SUMMARY

- A. Includes But Not Limited To:
 - 1. Killing and removing invasive plant material existing on site including, including but not limited to, Bermuda Grass.

1.2 RELATED SECTIONS

- A. Section 32 84 00 – Landscape Irrigation
- B. Section 32 93 00 – Landscape Plants
- C. Section 32 98 01 – Landscape Maintenance

1.3 DEFINITIONS

- A. Invasive Plants: Plant noted on the plans as not desirable and to be killed and removed.
- B. Landscape areas: All areas scheduled for turf, tree and shrub planting.
- C. Planters: All landscape areas not scheduled for turf.

1.4 QUALITY ASSURANCE

- A. Subcontractor license requirements are as noted in Section 32 9300, 1.7 Quality Assurance, under Installer.

1.5 SUBMITTALS

- A. Submit all MSDS sheets for herbicides, pre-emergent & post-emergent products that are used for this project.
- B. Submit a "Right to Know" form to the Owner and Principal (14) days before the application of herbicides, pre-emergent & post-emergent products, etc.

1.6 REGULATORY REQUIREMENTS

- A. Comply with regulatory requirements for herbicide composition.
- B. The types of herbicides to be used and the methods of application shall confirm with the Environmental Protection Agency Policies.

PART 2 - PRODUCTS

2.1 MANUFACTURES

- A. Post-Emergent products. Application rates as per manufacture recommendations.
- B. Post-Emergent, Round Up
- C. Post-Emergent, Fusilade
- D. Post-Emergent, Poast
- E. Post-Emergent, Diquat

PART 3 - EXECUTION

3.1 PERFORMANCE

- A. General
 - 1. Kill and then remove all Bermuda Grass in areas shown on the plans.
 - 2. Protect in place all existing plant material that is to remain. Damaged plant material that is to remain is to be replaced with a plant of the same genus, species, size and form as the damaged plant at the contractor's cost.
- B. All herbicide is to have a trace die coloring agent added for visual verification of application and coverage.

3.2 BERMUDA GRASS REMOVAL

- A. Bermuda grass removal during the dormant season when nighttime temperatures are consistently below 65 degrees Fahrenheit will consist of excavation only.
- B. Turf and Soil Removal
 - 1. Remove the turf base, roots and approximately 6"-7" of soil below the turf base depending on the thickness of the turf base for a total of 8" of material removed from the site.
 - 2. All removed Bermuda grass and soil is to be disposed of in a legal manner off site.
 - 3. Clean up all spillage, debris, roots, grass and other deleterious materials from the immediate working area and the surrounding site.
- C. Protection
 - 1. All trenching, digging, scraping or other work under limb spread of any and all evergreens or low branching deciduous material shall be done by hand or by other methods so as to prevent damage to limbs or branches.
 - 2. Where it is necessary to excavate adjacent to existing trees, use all possible care to avoid injury to trees and tree roots. Excavation, in areas where 2-inch and larger roots occur, shall be done by hand. Roots 2-inches or larger in diameter, except directly in the path of pipe or conduit, shall be tunneled under and shall be heavily wrapped with burlap to prevent scarring or excessive drying. Where a trenching machine is operated close to trees having roots smaller than 2-inches in diameter, a wall or trench adjacent to tree shall be hand trimmed, making clean cuts through roots. Roots 1-inch and larger in diameter shall be painted with two coats of "Tree Seal". Trenches adjacent to trees shall be closed within 24 hours, and when this is not possible, side of trench adjacent to tree shall be kept shaded with moistened burlap or canvas. Protect existing irrigation lines, wires, valves and other appurtenances that are to remain. Repair, replace or adjust any damaged irrigation equipment which is scheduled to remain in place. Make all repairs, replacements or adjustments immediately.
 - 3. Maintain continuous operation of existing irrigation system that is scheduled to remain during the entire duration of the scope of work. Immediately notify the A/E and NCSD Representative of any temporary water shut down or accidental damage.

D. Post Removal Herbicide Application

1. After removal of Bermuda grass and continuing through the maintenance period, the contractor will apply herbicide to any Bermuda grass that begins to grow in the areas where the Bermuda grass was excavated and removed.
2. Apply application of herbicide per manufacturer's written instructions.
 - a. Protect existing trees, shrubs and other plant material scheduled to remain from damage caused by the use of the herbicide on the Bermuda grass.
 - b. Use heaviest recommended application rate.
 - c. Thoroughly cover all green leaf blade surfaces of the Bermuda grass.
 - a. Apply herbicide to Bermuda grass that begins to grow at no more than two week intervals through the maintenance period.

3.3 FIELD QUALITY CONTROL

A. General

1. The A/E will inspect Invasive Plant Removal work regularly during the removal period.
2. Notify the A/E a minimum of 48 hours prior to each phase of work in this section.

END OF SECTION